CHAPTER
ONE
INTRODUCTION

1.1 CHARACTERIZATION OF ELECTROMAGNETIC RADIATION

Molecular spectroscopy may be defined as the study of the interaction of electromagnetic waves and matter. Throughout this book we shall be concerned with what spectroscopy can tell us of the structure of matter, so it is essential in this first chapter to discuss briefly the nature of electromagnetic radiation and the sort of interactions which may occur; we shall also consider, in outline, the experimental methods of spectroscopy.

Electromagnetic radiation, of which visible light forms an obvious but very small part, may be considered as a simple harmonic wave propagated from a source and travelling in straight lines except when refracted or reflected. The properties which undulate—corresponding to the physical displacement of a stretched string vibrating, or the alternate compressions and rarefactions of the atmosphere during the passage of a sound wave—are interconnected electric and magnetic fields. We shall see later that it is these undulatory fields which interact with matter giving rise to a spectrum.

It is trivial to show that any simple harmonic wave has properties of the sine wave, defined by $y = A \sin \theta$, which is plotted in Fig. 1.1. Here y is the displacement with a maximum value A,

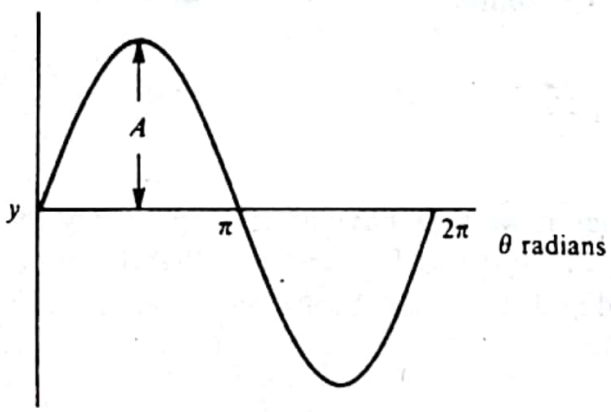


Figure 1.1 The curve of $y = A \sin \theta$.

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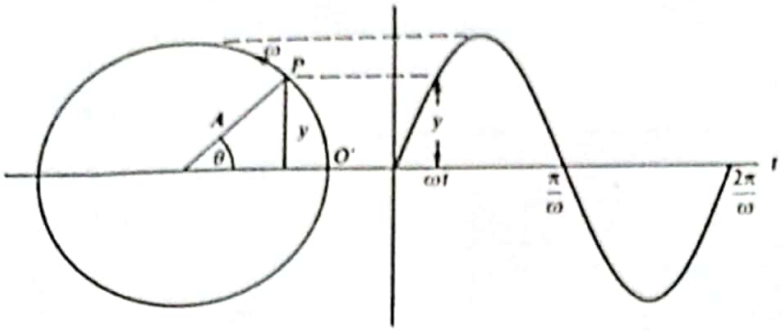


Figure 1.2 The description of a sine curve in terms of the circular motion of a point P at a uniform angular velocity of ω rad s^{-1}

and θ is an angle varying between 0 and 360° (or 0 and 2π radians). The relevance of this representation to a travelling wave is best seen by considering the left-hand side of Fig. 1.2. A point P travels with uniform angular velocity ω rad s^{-1} in a circular path of radius A; we measure the time from the instant when P passes O' and then, after a time t seconds, we imagine P to have described an angle $\theta = \omega t$ radians. Its vertical displacement is then $y = A \sin\theta = A \sin\omega t$, and we can plot this displacement against time as on the right-hand side of Fig. 1.2. After a time of $2\pi/\omega$ seconds, P will return to O', completing a 'cycle'. Further cycles of P will repeat the pattern and we can describe the displacement as a continuous function of time by the graph of Fig. 1.2.

In one second the pattern will repeat itself $\omega/2\pi$ times, and this is referred to as the *frequency* (ν) of the wave. The SI unit of frequency is called the hertz (abbreviated to Hz) and has the dimensions of reciprocal seconds (abbreviated s^{-1}). We may then write:

$$y = A \sin\omega t = A \sin 2\pi\nu t \tag{1.1}$$

as a basic equation of wave motion.

So far we have discussed the variation of displacement with time, but in order to consider the nature of a *travelling* wave, we are more interested in the distance variation of the displacement. For this we need the fundamental distance–time relationship:

$$x = ct \tag{1.2}$$

where x is the distance covered in time t at a speed c. Combining (1.1) and (1.2) we have:

$$y = A \sin 2\pi\nu t = A \sin \frac{2\pi\nu x}{c}$$

and the wave is shown in Fig. 1.3. Besides the frequency ν, we now have another property by which we can characterize the wave—its *wavelength* λ, which is the distance travelled during a complete cycle. When the velocity is c metres per second and there are ν cycles per second, there are evidently ν waves in c metres, or

$$\nu\lambda = c \qquad \lambda = c/\nu \quad \text{metres} \tag{1.3}$$

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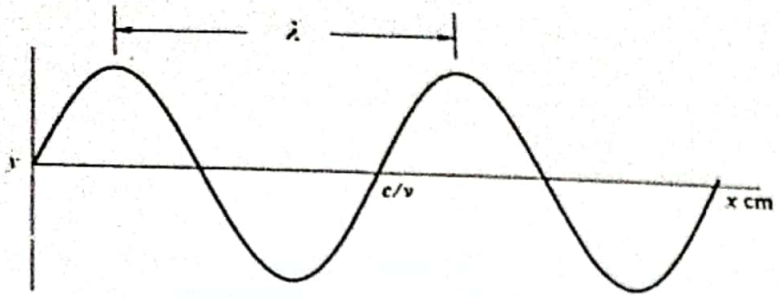


Figure 1.3 The concept of a travelling wave of wavelength λ.

so we have:

$$y = A \sin \frac{2\pi x}{\lambda} \tag{1.4}$$

In spectroscopy wavelengths are expressed in a variety of units, chosen so that in any particular range (see Fig. 1.4) the wavelength does not involve large powers of ten. Thus, in the microwave region, λ is measured in centimetres or millimetres, while in the infra-red it is usually given in micrometres (μm)—formerly called the *micron*—where:

$$1\ \mu\text{m} = 10^{-6}\ \text{m} \tag{1.5}$$

In the visible and ultra-violet region λ is usually expressed in nanometres (10^{-9} m), although occasional use is still made of the non-SI ångström unit where:

$$1\ \text{Å} = 10^{-10}\ \text{m} \qquad \text{or} \qquad 1\ \text{nm} = 10^{-9}\ \text{m} = 10\ \text{Å} \tag{1.6}$$

There is yet a third way in which electromagnetic radiation can be usefully characterized, and this is in terms of the *wavenumber* $\bar{\nu}$. Formally this is defined as the reciprocal of the wavelength expressed in *centimetres*:

$$\bar{\nu} = 1/\lambda\ \text{cm}^{-1} \tag{1.7}$$

and hence

$$y = A \sin 2\pi\bar{\nu}x \tag{1.8}$$

It is more useful to think of the wavenumber, however, as the number of complete waves or cycles contained in each centimetre length of radiation.

It is unfortunate that the conventional symbols of wavenumber ($\bar{\nu}$) and frequency (ν) are similar; confusion should not arise, however, if the units of any expression are kept in mind, since wavenumber is invariably expressed in reciprocal centimetres (cm^{-1}) and frequency in cycles per second (s^{-1} or Hz). The two are, in fact, proportional: $\nu = c\bar{\nu}$, where the proportionality constant is the velocity of radiation expressed in *centimetres* per second (that is $3 \times 10^{10}\ \text{cm s}^{-1}$).

1.2 THE QUANTIZATION OF ENERGY

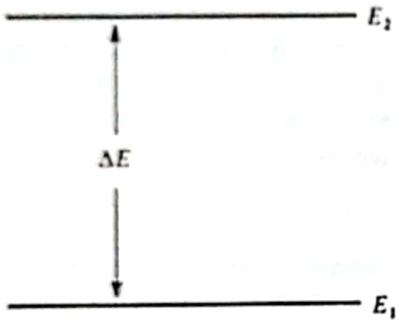
Towards the end of the last century experimental data were observed which were quite incompatible with the previously accepted view that matter could take up energy continuously. In 1900 Max Planck published the revolutionary idea that the energy of an oscillator is discontinuous and that any change in its energy content can occur only by means of a jump between

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two distinct energy states. The idea was later extended to cover many other forms of the energy of matter.

A molecule in space can have many sorts of energy; e.g. it may possess rotational energy by virtue of bodily rotation about its centre of gravity; it will have vibrational energy due to the periodic displacement of its atoms from their equilibrium positions; it will have electronic energy since the electrons associated with each atom or bond are in unceasing motion, etc. The chemist or physicist is early familiar with the electronic energy states of an atom or molecule and accepts the idea that an electron can exist in one of several discrete energy levels: he learns to speak of the energy as being *quantized*. In much the same way the rotational, vibrational, and other energies of a molecule are also quantized—a particular molecule can exist in a variety of rotational, vibrational, etc., energy levels and can move from one level to another only by a sudden jump involving a finite amount of energy.

Consider two possible energy states of a system—two rotational energy levels of a molecule, for example—labelled E_1 and E_2 in the following diagram:



The suffixes 1 and 2 used to distinguish these levels are, in fact, *quantum numbers*. The actual significance of quantum numbers goes far deeper than their use as a convenient label—in particular, we shall later see that analytical expressions for energy levels usually involve an algebraic function of one or more quantum numbers. Transitions can take place between the levels E_1 and E_2 provided the appropriate amount of energy, $\Delta E = E_2 - E_1$, can be either absorbed or emitted by the system. Planck suggested that such absorbed or emitted energy can take the form of electromagnetic radiation and that the frequency of the radiation has the simple form:

$$\nu = \Delta E/h \quad \text{Hz}$$

i.e.

$$\Delta E = h\nu \quad \text{joules} \tag{1.9}$$

where we express our energies E in terms of the joule, and h is a universal constant—Planck's constant. This suggestion has been more than amply confirmed by experiment.

The significance of this is that if we take a molecule in state 1 and direct on to it a beam of radiation of a single frequency ν (monochromatic radiation), where $\nu = \Delta E/h$, energy will be absorbed from the beam and the molecule will jump to state 2. A detector placed to collect the radiation after its interaction with the molecule will show that its intensity has decreased. Also if we use a beam containing a wide range of frequencies ('white' radiation), the detector will show that energy has been absorbed *only* from that frequency $\nu = \Delta E/h$, all other frequencies being undiminished in intensity. In this way we have produced a spectrum—an *absorption* spectrum.

Alternatively the molecule may already be in state 2 and may revert to state 1 with the consequent emission of radiation. A detector would show this radiation to have frequency

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$\nu = \Delta E/h$ only, and the *emission* spectrum so found is plainly complementary to the absorption spectrum of the previous paragraph.

The actual energy differences between the rotational, vibrational, and electronic energy levels are very small and may be measured in joules per molecule (or atom). In these units Planck's constant has the value:

$$h = 6.63 \times 10^{-34} \text{ joules s molecule}^{-1}$$

Often we are interested in the total energy involved when a gram-molecule of a substance changes its energy state: for this we multiply by the Avogadro number $N = 6.02 \times 10^{23}$.

However, the spectroscopist measures the various characteristics of the absorbed or emitted radiation during transitions between energy states and often, rather loosely, uses frequency, wavelength, and wavenumber as if they were energy units. Thus in referring to 'an energy of $10\,\text{cm}^{-1}$', the spectroscopist means 'a separation between two energy states such that the associated radiation has a wavenumber value of $10\,\text{cm}^{-1}$'. The first expression is so simple and convenient that it is essential to become familiar with wavenumber and frequency energy units if one is to understand the spectroscopist's language. Throughout this book we shall use the symbol ε to represent energy in cm^{-1}.

It cannot be too firmly stressed at this point that the frequency of radiation associated with an energy change does *not* imply that the transition between energy levels occurs a certain number of times each second. Thus an electronic transition in an atom or molecule may absorb or emit radiation of frequency some 10^{15} Hz, but the electronic transition does not itself *occur* 10^{15} times per second. It may occur once or many times and on each occurrence it will absorb or emit an energy quantum of the appropriate frequency.

1.3 REGIONS OF THE SPECTRUM

Figure 1.4 illustrates in pictorial fashion the various, rather arbitrary, regions into which electromagnetic radiation has been divided. The boundaries between the regions are by no means precise, although the molecular processes associated with each region are quite different. Each succeeding chapter in this book deals essentially with one of these processes.

In increasing frequency the regions are:

1. Radiofrequency region: 3×10^6–3×10^{10} Hz; 10 m–1 cm wavelength. Nuclear magnetic resonance (n.m.r.) and electron spin resonance (e.s.r.) spectroscopy. The energy change involved is that arising from the reversal of spin of a nucleus or electron, and is of the order 0.001–10 joules/mole (Chapter 7).
2. Microwave region: 3×10^{10}–3×10^{12} Hz; 1 cm–100 μm wavelength. Rotational spectroscopy. Separations between the rotational levels of molecules are of the order of hundreds of joules per mole (Chapter 2).
3. Infra-red region: 3×10^{12}–3×10^{14} Hz; 100 μm–1 μm wavelength. Vibrational spectroscopy. One of the most valuable spectroscopic regions for the chemist. Separations between levels are some 10^4 joules/mole (Chapter 3).
4. Visible and ultra-violet regions: 3×10^{14}–3×10^{16} Hz; 1 μm–10 nm wavelength. Electronic spectroscopy. The separations between the energies of valence electrons are some hundreds of kilojoules per mole (Chapters 5 and 6).
5. X-ray region: 3×10^{16}–3×10^{18} Hz; 10 nm–100 pm wavelength. Energy changes involving the inner electrons of an atom or a molecule, which may be of order ten thousand kilojoules (Chapter 5).

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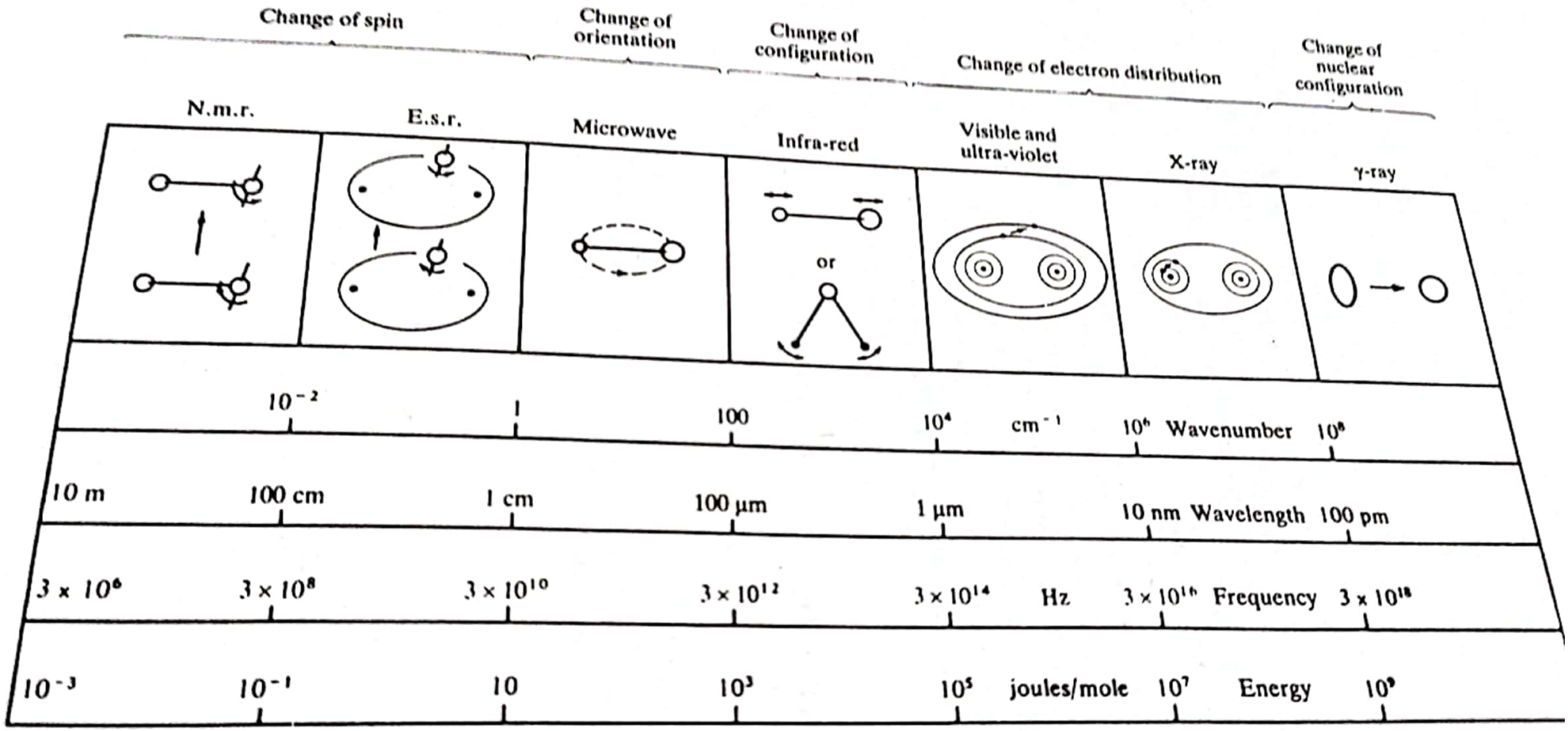


Figure 1.4 The regions of the electromagnetic spectrum.

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6. γ-ray region: 3×10^{18}–3×10^{20} Hz; 100 pm–1 pm wavelength. Energy changes involve the rearrangement of nuclear particles, having energies of 10^9–10^{11} joules per gram atom (Chapter 9).

One other type of spectroscopy, that discovered by Raman and bearing his name, is discussed in Chapter 4. This, it will be seen, yields information similar to that obtained in the microwave and infra-red regions, although the experimental method is such that observations are made in the visible region.

In order that there shall be some mechanism for interaction between the incident radiation and the nuclear, molecular, or electronic changes depicted in Fig. 1.4, there must be some electric or magnetic effect produced by the change which can be influenced by the electric or magnetic fields associated with the radiation. There are several possibilities:

1. The radiofrequency region. We may consider the nucleus and electron to be tiny charged particles, and it follows that their spin is associated with a tiny magnetic dipole. The reversal of this dipole consequent upon the spin reversal can interact with the magnetic field of electromagnetic radiation at the appropriate frequency. Consequently all such spin reversals produce an absorption or emission spectrum.
2. The microwave region. A molecule such as hydrogen chloride, HCl, in which one atom (the hydrogen) carries a permanent net positive charge and the other a net negative charge, is said to have a permanent electric dipole moment. H_2 or Cl_2, on the other hand, in which there is no such charge separation, have a zero dipole. If we consider the rotation of HCl (Fig. 1.5, where we notice that if only a pure rotation takes place, the centre of gravity of the molecule must not move), we see that the plus and minus charges change places periodically, and the component dipole moment in a given direction (say upwards in the plane of the paper) fluctuates regularly. This fluctuation is plotted in the lower half of Fig. 1.5, and it is seen to be exactly similar in form to the fluctuating electric field of radiation (cf. Fig. 1.2). Thus interaction can occur, energy can be absorbed or emitted, and the rotation gives rise to a spectrum. All molecules having a permanent moment are said to be 'microwave active'. If there is no dipole, as in H_2 or Cl_2, no interaction can take place and the molecule is 'microwave inactive'. This imposes a limitation on the applicability of microwave spectroscopy.

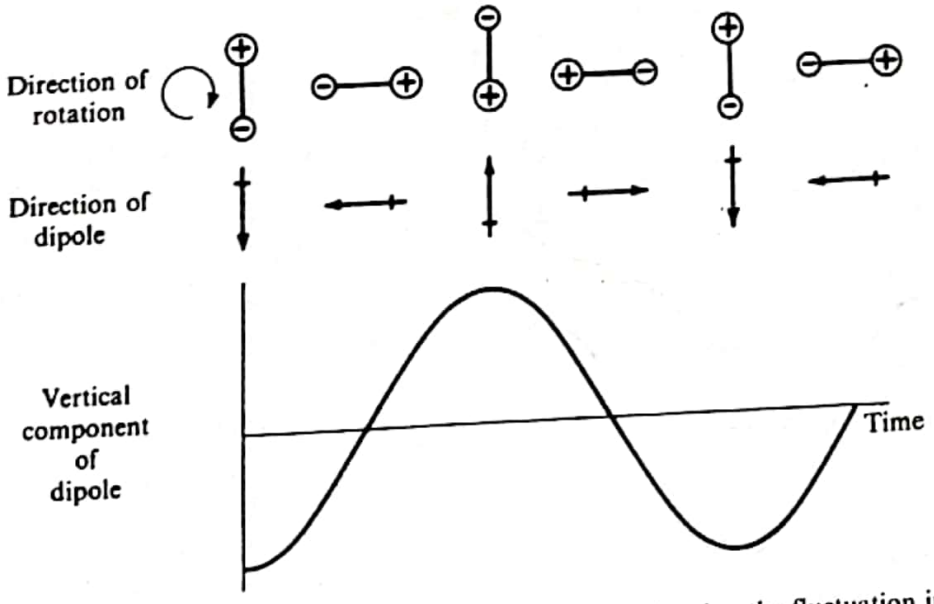


Figure 1.5 The rotation of a polar diatomic molecule, showing the fluctuation in the dipole moment measured in a particular direction.

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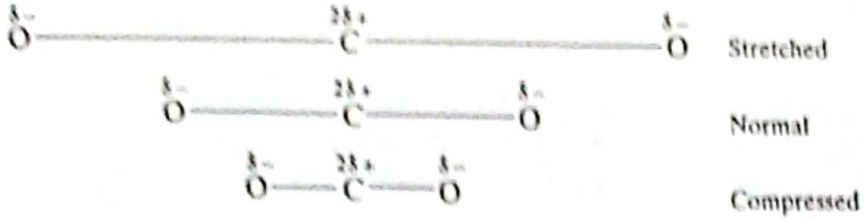
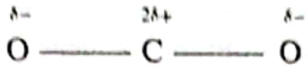


Figure 1.6 The symmetric stretching vibration of the carbon dioxide molecule with amplitude much exaggerated.

3. The infra-red region. Here it is a vibration, rather than a rotation, which must give rise to a dipole change. Consider the carbon dioxide molecule as an example, in which the three atoms are arranged linearly with a small net positive charge on the carbon and small negative charges on the oxygens:



During the mode of vibration known as the 'symmetric stretch', the molecule is alternately stretched and compressed, both C—O bonds changing simultaneously, as in Fig. 1.6. Plainly the dipole moment remains zero throughout the whole of this motion, and this particular vibration is thus 'infra-red inactive'.

However, there is another stretching vibration called the anti-symmetrical stretch, depicted in Fig. 1.7. Here one bond stretches while the other is compressed, and vice versa. As the

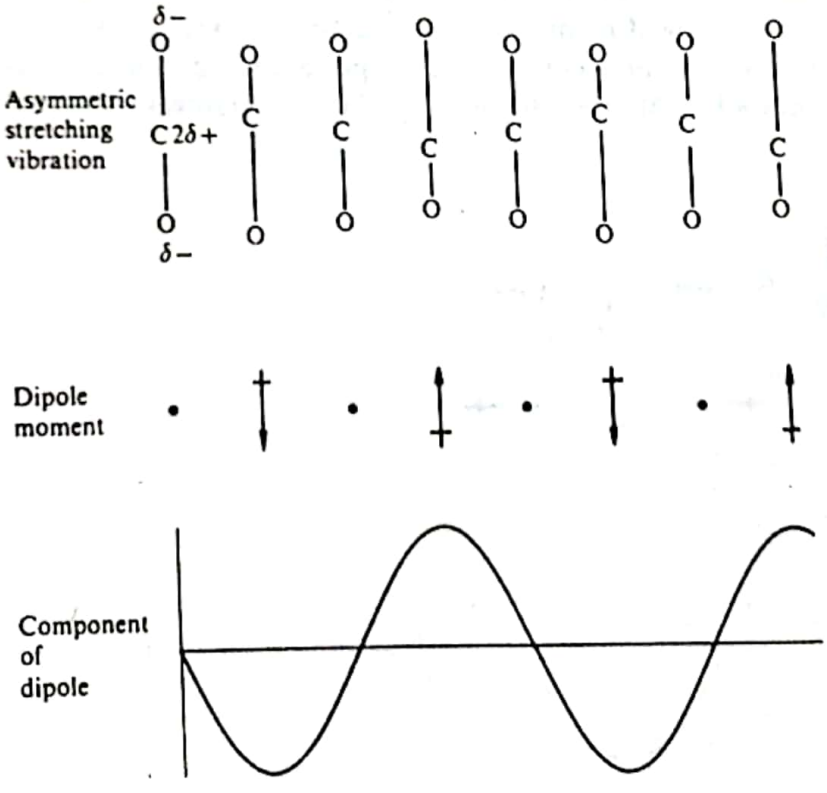


Figure 1.7 The asymmetric stretching vibration of the carbon dioxide molecule, showing the fluctuation in the dipole moment.

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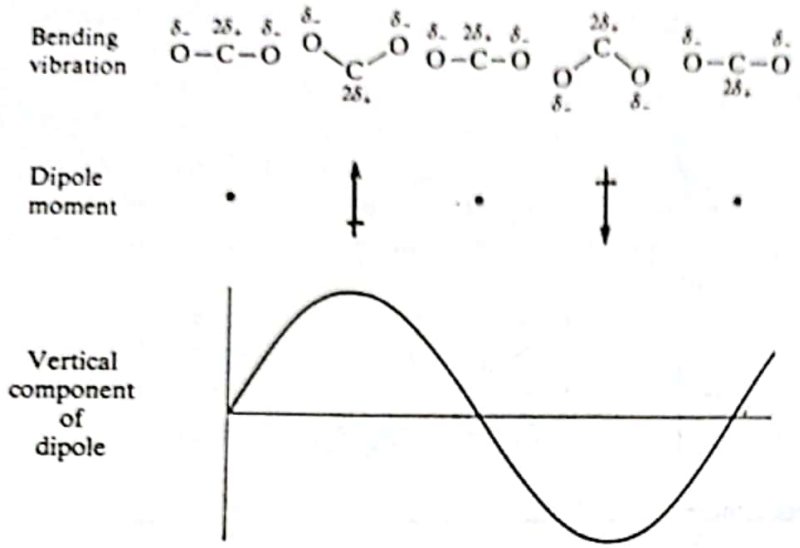


Figure 1.8 The bending motion of the carbon dioxide molecule and its associated dipole fluctuation.

figure shows, there is a periodic alteration in the dipole moment, and the vibration is thus 'infra-red active'. One further vibration is allowed to this molecule (see Chapter 3 for a more detailed discussion), known as the bending mode. This, as shown in Fig. 1.8, is also infra-red active. In neither of these motions does the centre of gravity move. Note particularly that the relative motions of the atoms are very much exaggerated in Figs 1.6, 1.7, and 1.8; in real molecules the displacement of atoms during vibrations is seldom more than about 10 per cent of the bond length.

Although dipole change requirements do impose some limitation on the application of infra-red spectroscopy, the appearance or non-appearance of certain vibration frequencies can give valuable information about the structure of a particular molecule (see Chapter 3).

4. The visible and ultra-violet region. The excitation of a valence electron involves the moving of electronic charges in the molecule. The consequent change in the electric dipole gives rise to a spectrum by its interaction with the undulatory electric field of radiation.
5. There is a rather special requirement for a molecular motion to be 'Raman active'; this is that the electrical *polarizability* of the molecule must change during the motion. This will be discussed fully in Chapter 4.

1.4 REPRESENTATION OF SPECTRA

We show in Fig. 1.9 a highly schematic diagram of a recording spectrometer suitable for use in the ultra-violet, visible and infra-red regions of the spectrum; since it uses a grating (a block of reflective material with a grid of parallel lines ruled on its surface) to select the frequencies which pass through, it is usually called a 'grating spectrometer'; another term frequently used is 'dispersive spectrometer' since the grating 'disperses' the radiation into its frequency components.

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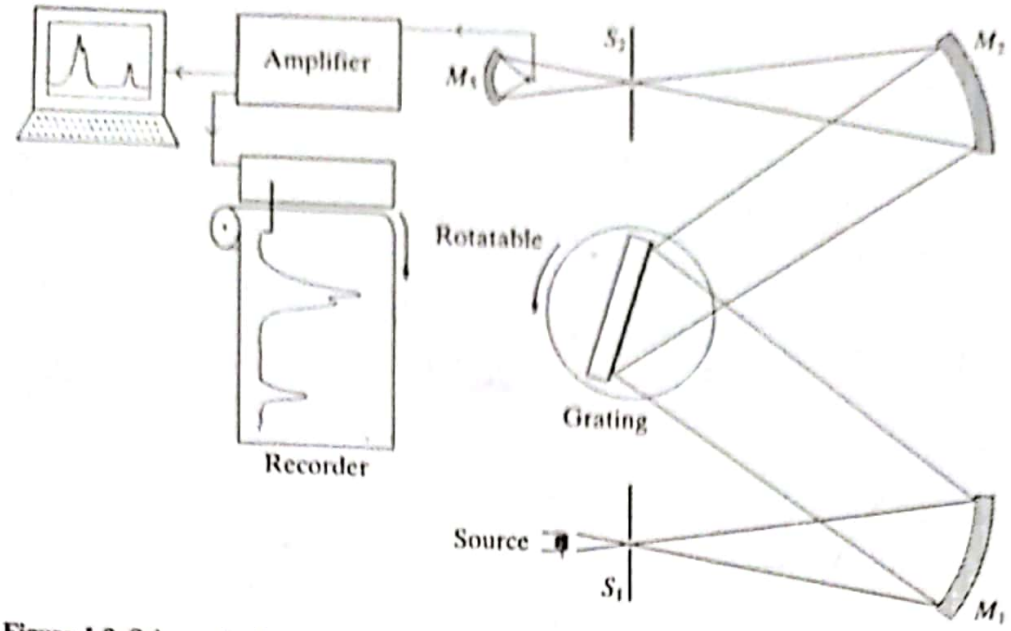


Figure 1.9 Schematic diagram of a grating spectrometer.

Radiation from the source, shown here as a hot or incandescent filament, passes through the vertical entrance slit S_1 (seen from above in the diagram) on to the spherical mirror M_1, from which a parallel beam is reflected on to the grating. The source is 'white', i.e. it radiates energy over a wide range of frequencies and, because of interference at the grid of the grating, different frequencies reflect from the grating at different angles. Thus when the grating is rotated as shown, a succession of frequencies is swept across the mirror M_2, from where it is focused on to the exit slit, S_2. So we see that the frequency of the radiation arriving at S_2 depends on the angle of the grating.

From S_2 the radiation falls on M_3, by which it is focused on to the detector. The latter, perhaps a thermocouple for infra-red radiation or a photomultiplier for visible and ultra-violet, responds with an electrical output proportional to the intensity of the radiation falling on it. This signal is amplified electronically, and then used either to drive a pen, which records the spectrum immediately, or is collected and stored on a computer for later processing and display.

The purpose of the slits is threefold. Firstly, they provide a sharply defined image which can be focused on the detector; secondly, the narrower they are the smaller is the frequency range which passes through, and consequently the resolving power (see Sec. 1.6) of the instrument is increased; and finally they help to prevent stray radiation, perhaps scattered from various components of the spectrometer or arriving from outside, getting to the detector and thus being falsely recorded as part of the spectrum.

The sample (not shown in the diagram) can be placed almost anywhere in the radiation beam. In order to be able to minimize its size, it is often placed close to one of the slits, where the beam is smallest; also it is preferable to put it near the exit slit, S_2, where it is bathed in only a narrow range of frequencies at any one time, rather than near S_1 where it is subjected to the full output of the source and may suffer some degradation from the intense heat or light.

Let us consider now what happens when a spectrum is recorded. If there is no sample present the detector output will, ideally, be constant over the range of frequencies covered by the instrument. On the other hand, if we imagine putting into the beam a substance having just two possible energy levels, E_1 and E_2, the detector output will show a sudden fall at a frequency given by $\nu = (E_2 - E_1)/h$, since some energy at this frequency will be absorbed by the sample and will no longer reach the detector. The resulting trace on the chart paper is illustrated in Fig. 1.10(a), where we show, on the left, the detector output (with 100 per cent at the bottom) and,

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on the right, the percentage of the energy absorbed by the sample. We would say that the spectrum has been *scanned* between the beginning and ending frequencies, and such a picture is referred to, rather grandly, as a spectrum in the 'frequency domain', to indicate that it records detector output against frequency. In Section 1.8 we shall discuss 'time domain' spectroscopy, where the detector output is recorded as a function of time.

The ideal situation of Fig. 1.10(*a*) is seldom attained. Not only does the source emissivity vary with frequency, but often the sensitivity of the detector is also frequency-dependent. Thus the baseline—the no-sample condition—is seldom horizontal, although matters can usually be arranged so that it is approximately linear. Further, since it is impossible to make the slits infinitely narrow, a *range* of frequencies, rather than just a single frequency, falls on the detector at any given position of the grating. Additionally, we shall see in Sec. 1.7 that no energy transitions in atoms and molecules are absolutely sharp, but they always occur over a range of frequencies. These factors cause a broadening of the spectral absorption and Fig. 1.10(*b*) illustrates a more typical spectrum showing their effects. Also shown are the small peaks and troughs arising from randomly generated electrical signals in the detector, the amplifier, or the pen recorder, usually collectively referred to as 'noise' on the spectrum.

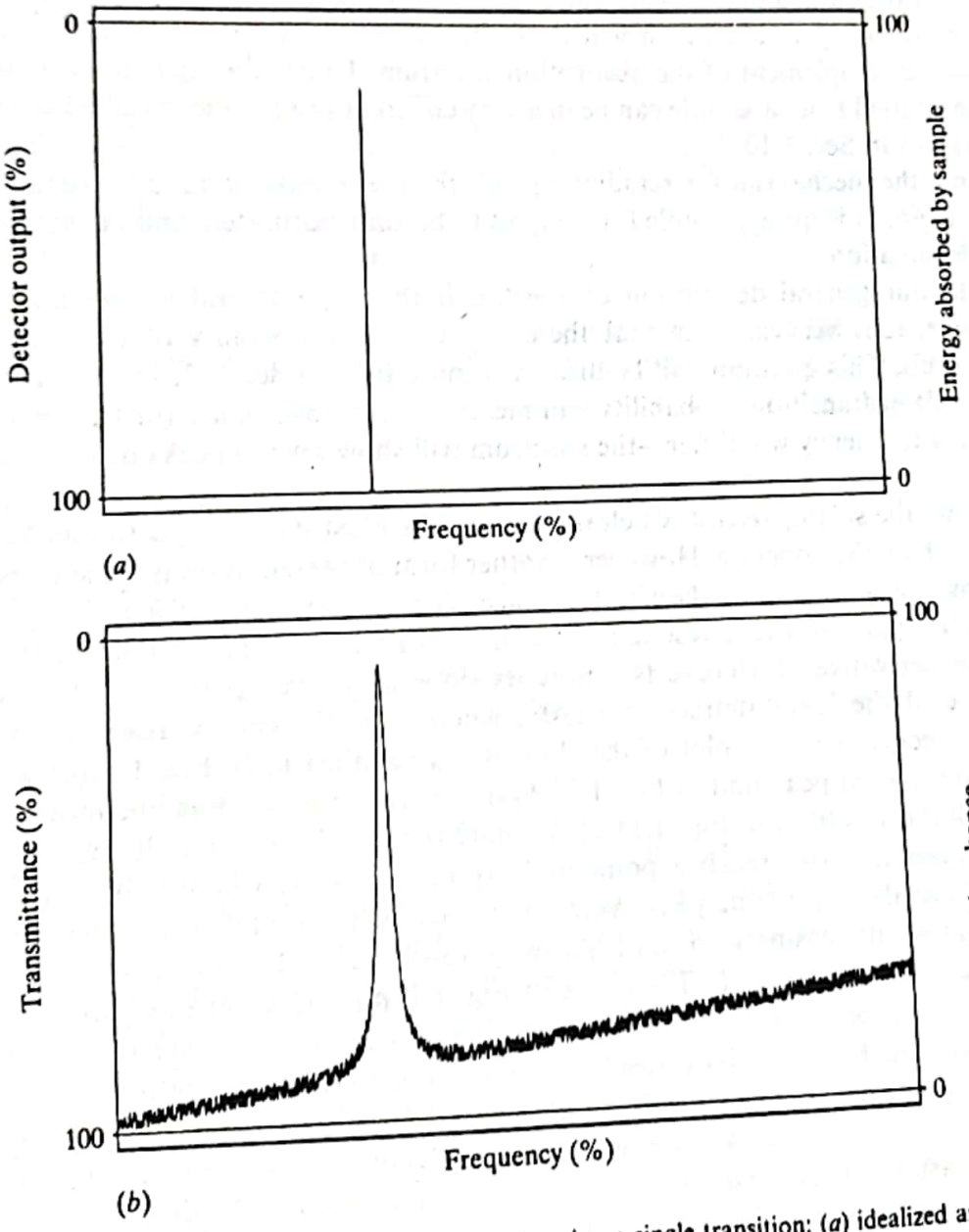


Figure 1.10 The spectrum of a molecule undergoing a single transition: (*a*) idealized and (*b*) usual appearance.

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At this point it is helpful to pause briefly in our discussion of spectra and consider what happens to the energy absorbed in the sort of process discussed above. In the ultra-violet, visible, and infra-red regions it is an experimental fact that a given sample continues to show an absorption spectrum for as long as we care to irradiate it—in other words a finite number of sample molecules appear to be capable of absorbing an infinite amount of energy. Plainly the molecules must be able to rid themselves of the absorbed energy. The most common mechanism for this is thermal collisions. An energized molecule collides with its neighbours and gradually loses its excess energy to them as kinetic energy—the sample as a whole becomes warm.

Another mechanism is that energy gained from radiation is lost as radiation once more. A molecule in the ground state absorbs energy at frequency ν and its energy is raised an amount $\Delta E = h\nu$ above the ground state. It is thus in an excited, unstable, condition, but by emitting radiation of frequency ν again, it can revert to the ground state and is able to reabsorb radiation from the source beam once more. In such a case it may be asked how an absorption spectrum can arise at all, since the absorbed energy is re-emitted by the sample. The answer is simply that the radiation is re-emitted in a random direction and the proportion of such radiation reaching the detector is tiny—in fact re-emitted radiation has the same chance of reaching the source as the detector. The net effect, then, is an absorption from the directed beam and, when re-emission occurs, a scattering into the surroundings. The scattered radiation can, of course, be collected and observed as an emission spectrum which will be—with important reservations to be discussed in Chapter 4—the complement of the absorption spectrum. Under the right conditions much of the radiation emitted from a sample can be in a very coherent beam—the so-called laser radiation. We discuss this in Sec. 1.10.

On the other hand, the mechanism for reradiating *radiofrequency* radiation is not particularly efficient. In this region it is quite possible for samples to become 'saturated' and be unable to continue to absorb radiation.

Returning now to our general description of spectra: if there are several energy levels available to the sample, it is very unlikely that there is the same probability of transition between the various levels. This question will be discussed more fully in Sec. 1.7, but we may here note that differences in transition probability will mean that the absorbance (or transmittance) at each absorbing frequency will differ—the spectrum will show several peaks of varying height.

Figure 1.10(*b*) shows the sort of record which is produced by most modern spectrometers, whatever the region in which they operate. However, another form of presentation is sometimes adopted—that showing the *derivative* of the spectral trace. A prime example of a technique using this form is electron paramagnetic resonance which shows relatively broad lines in the microwave region. The derivative of a curve is simply its slope at a given point; in calculus notation, the derivative of the spectral trace is $dA/d\nu$, where A is the energy absorbed or emitted. The derivative record is thus a plot of the slope $dA/d\nu$ against ν. In Fig. 1.11(*a*) we show a typical broad absorption peak and in Fig. 1.11(*b*) its derivative curve. Imagine moving from left to right across the spectrum in Fig. 1.11(*a*); we note that its slope is initially positive and small, that it increases until we reach a point midway up the peak, where it begins to decrease, becoming zero at the top of the peak, where the trace is horizontal. From then on the slope is always negative, increasingly so until half-way down the peak, and then steadily decreasing until it eventually becomes zero. The curve of Fig. 1.11(*b*) traces out this pattern.

Although at first sight more complex, the derivative trace has advantages over the direct record in some circumstances. Firstly, it indicates rather more precisely the centre of a broad absorbance peak; at this point the A curve is horizontal, hence $dA/d\nu$ is zero, and the centre is marked by the intersection of the derivative curve with the axis. Secondly, the width of a peak is often more accurately measured from the derivative curve. The width is normally defined as the 'half-width', i.e. the width of the absorbance peak at half its height, and this, as we can see from

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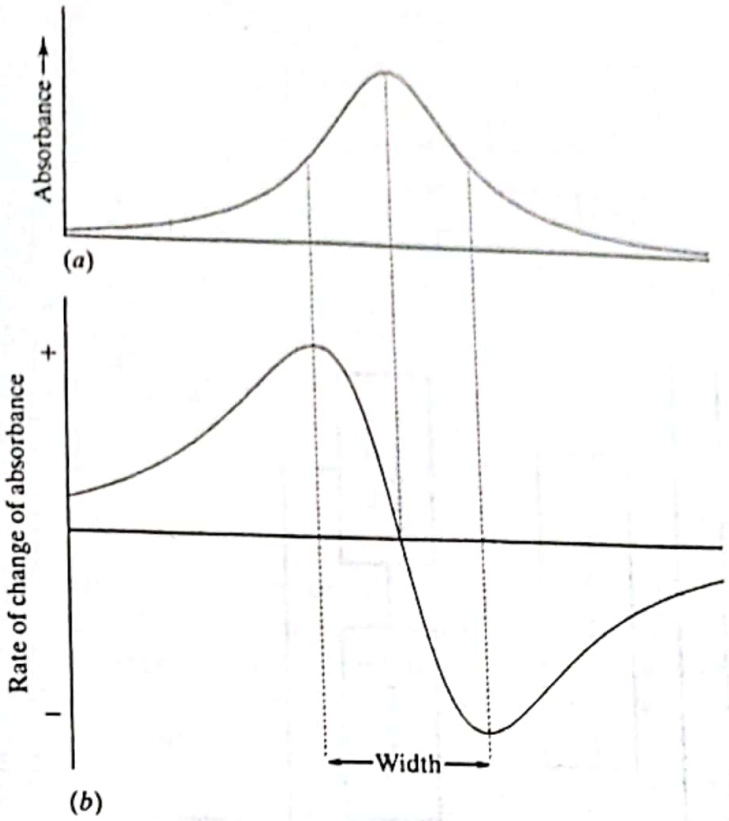


Figure 1.11 The relationship between absorption and derivative spectra: (*a*) a broad absorption band and (*b*) its derivative.

Fig. 1.11(*b*), is shown very clearly by the frequency difference between the maximum and the minimum of the derivative. Finally, for instrumental reasons, it is often better to measure the relative intensities of absorptions from the derivative curve rather than the direct trace.

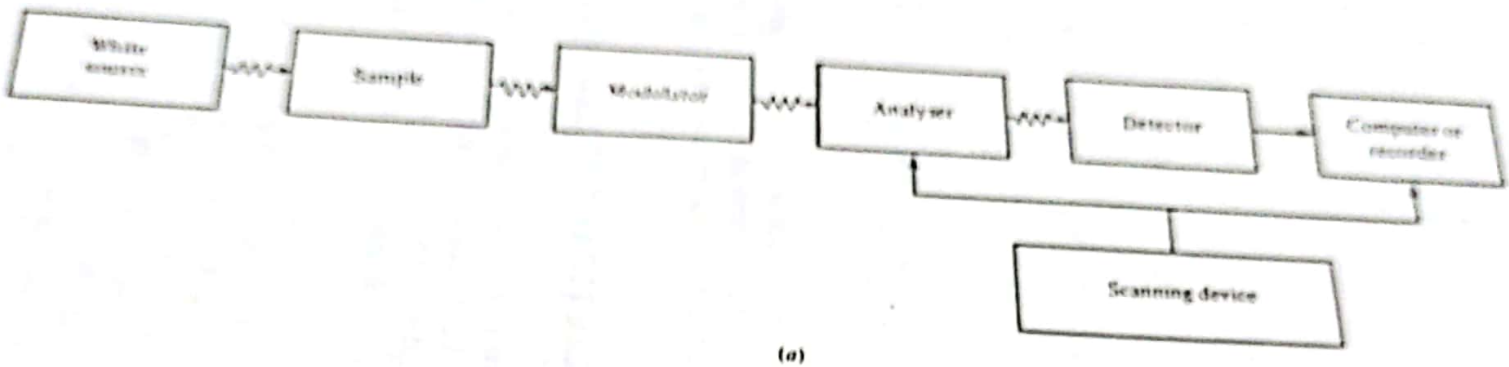
1.5 BASIC ELEMENTS OF PRACTICAL SPECTROSCOPY

Spectrometers used in various regions of the spectrum naturally differ widely from each other in construction. These differences will be discussed in more detail in the following chapters, but here it will probably be helpful to indicate the basic features which are common to all types of spectrometer. We may, for this purpose, consider absorption and emission spectrometers separately.

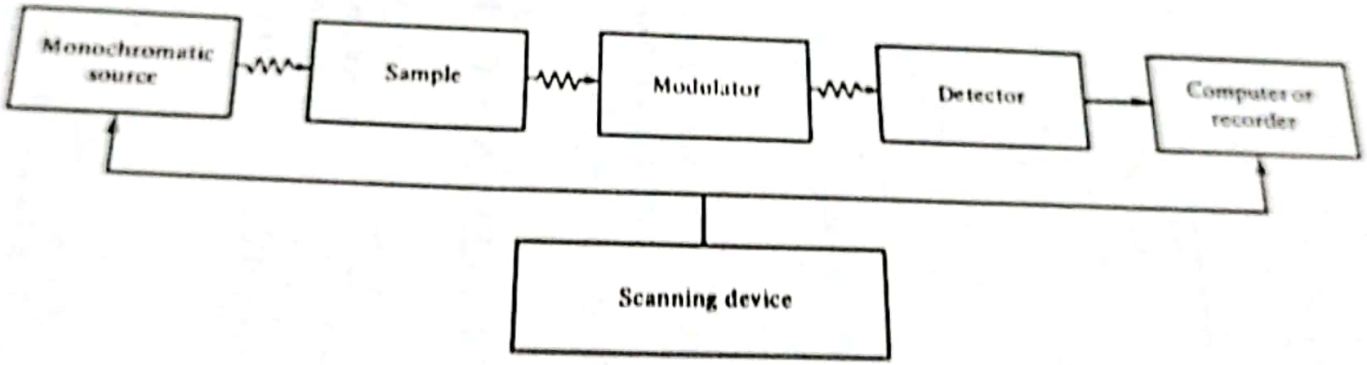
1. *Absorption instruments.* Figure 1.12(*a*) shows, in block diagram form, the components of an absorption spectrometer which might be used in the infra-red, visible, and ultra-violet regions. The radiation from a white source is directed by some guiding device (e.g. the mirror of Fig. 1.9) on to the sample, from which it passes through an analyser (e.g. the grating of Fig. 1.9), which selects the frequency reaching the detector at any given time. The signal from the latter passes to a recorder which is synchronized with the analyser so as to produce a trace of the absorbance as the frequency varies.

   Placed, often, between the sample and the analyser is a *modulator*; this mechanical or electronic device interrupts the radiation beam a certain number of times per second, usually fixed somewhere between 10 and 1000 times, and its effect is to cause the detector to send an alternating current signal to the recorder, with a fixed frequency of 10–1000 Hz, rather than

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(*a*)



(*b*)

Figure 1.12 *Block diagram of a typical scanning absorption spectrometer for use in (a) the infra-red, visible and ultra-violet regions, where a 'white' source is available, and (b) the microwave and radiofrequency regions where the source can be tuned over a range of frequencies.*

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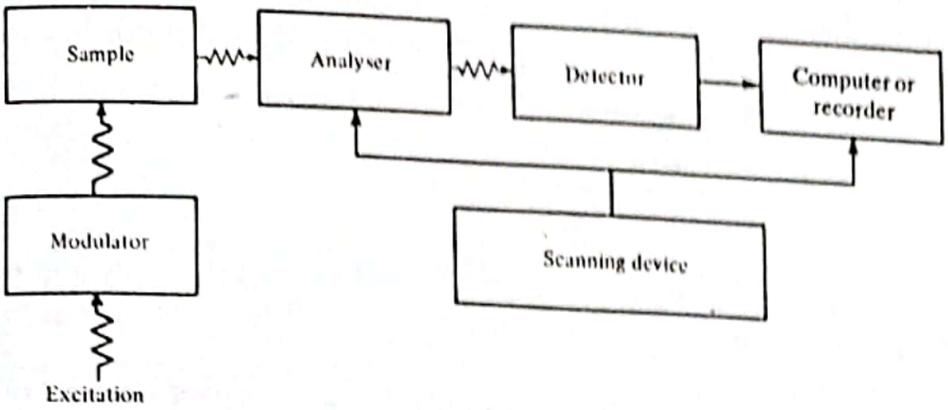


Figure 1.13 Block diagram of a typical emission spectrometer.

the direct current signal which would result from a steady, uninterrupted beam. This has two main advantages: (*a*) the amplifier in the recorder can be of a.c. type which is, in general, simpler to construct and more reliable in operation than a d.c. amplifier, and (*b*) the amplifier can be tuned to select only that frequency which the modulator imposes on the signal, thus ignoring all other signals. In this way stray radiation and other extraneous signals are removed from the spectral trace and a better, cleaner spectrum results.

In the microwave and radiofrequency regions it is possible to construct monochromatic sources whose emission frequency can be varied over a range. In this case, as Fig. 1.12(*b*) shows, no analyser is necessary, the source being, in a sense, its own analyser. Now it is necessary for the recorder to be synchronized with the source-scanning device in order that a spectral trace be obtained.

2. *Emission instruments.* The layout now differs in that the sample, after excitation, is its own source, and it is necessary only to collect the emitted radiation, analyse, and record it in the usual way. Figure 1.13 shows, schematically, a typical spectrometer. The excitation can be thermal or electrical, but often takes the form of electromagnetic radiation. In the latter case it is essential that the detector does not collect radiation directly from the exciting beam, and the two are placed at right angles as shown. A modulator placed between the source of excitation and the sample, together with a tuned detector-amplifier, ensures that the only emission recorded from the sample arises directly from excitation; any other spontaneous emission is ignored.

1.6 SIGNAL-TO-NOISE: RESOLVING POWER

Two other spectroscopic terms may be conveniently discussed at this point since they will recur in succeeding chapters.

1.6.1 Signal-to-Noise Ratio

Since almost all modern spectrometers use some form of electronic amplification to magnify the signal produced by the detector, every recorded spectrum has a background of random fluctuations caused by spurious electronic signals produced by the source or detector, or generated in the amplifying equipment. These fluctuations are usually referred to as 'noise'. In order that a real spectral peak should show itself as such and be sufficiently distinguished from the noise, it must have an intensity some three or four times that of the noise fluctuations (a signal-to-noise

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ratio of three or four). This requirement places a lower limit on the intensity of observable signals. In Sec. 1.9 we refer briefly to a computer-averaging technique by which it is possible to improve the effective signal-to-noise ratio.

1.6.2 Resolving Power

This is a somewhat imprecise concept which can, however, be defined rather arbitrarily and is often used as a measure of the performance of a spectrometer. We shall here consider it in general terms only.

No molecular absorption takes place at a single frequency only, but always over a spread of frequencies, usually very narrow but sometimes quite large (see Sec. 1.7); it is for this reason that we have up to now drawn spectra with broadened line shapes.

Let us consider two such lines close together, as on the right of Fig. 1.14(*a*): the dotted curve represents the absorption due to each line separately, the full line their combined absorption. We shall first take the exit slit width to be larger than the separation between the lines. Scanning the spectrum plainly involves moving the twin absorbance peaks steadily to the left so that they pass across the exit slit and into the detector; the situation at successive stages is shown in (*b*), (*c*), and (*d*) of Fig. 1.14, the shaded area showing the amount of absorbance which the detector

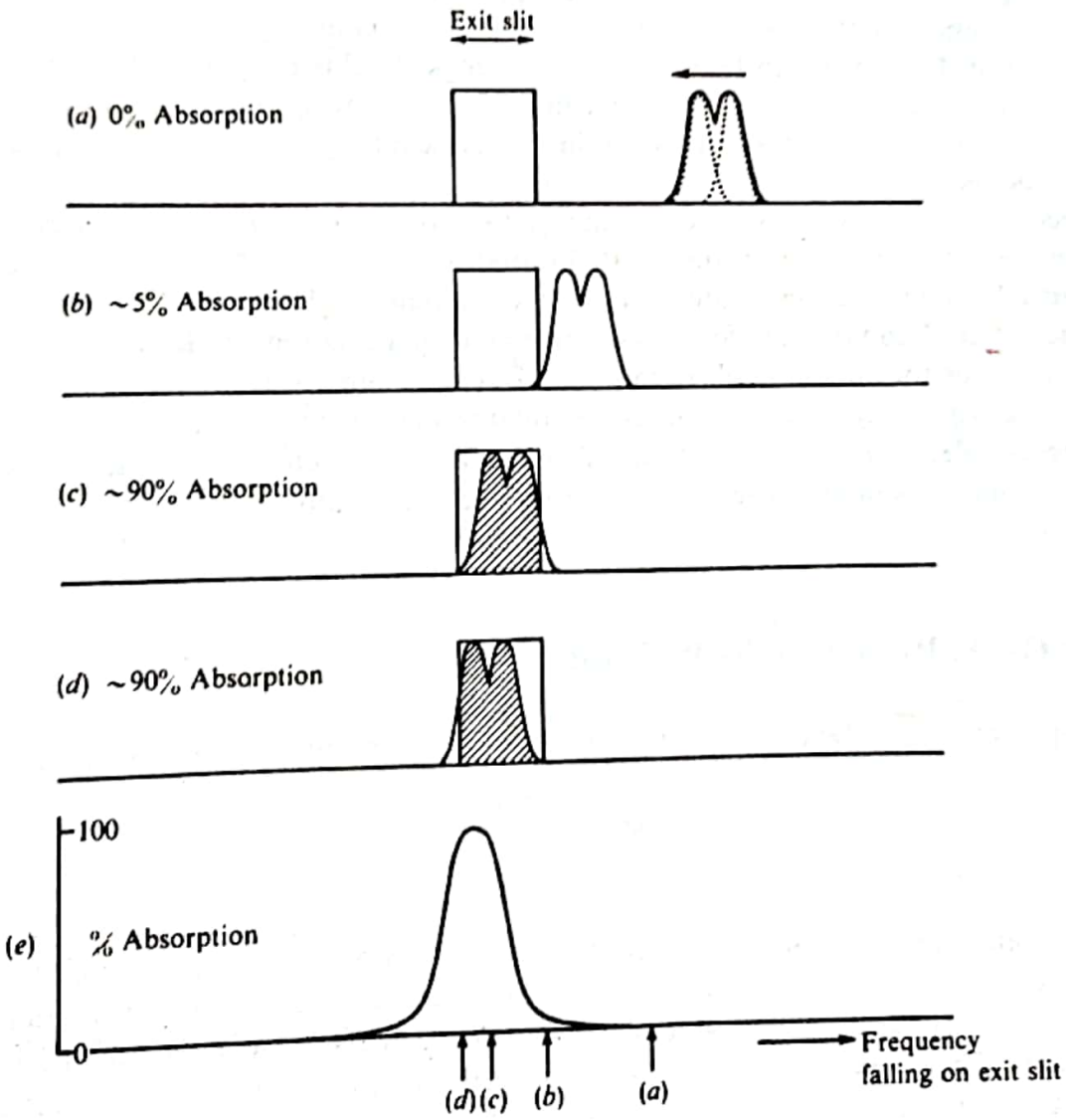


Figure 1.14 Illustrating the relationship between slit width and resolving power.

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would register. At (*e*) of this figure, the absorbance is plotted against frequency, together with the approximate positions of stages (*a*), (*b*), (*c*), and (*d*).

It is quite evident that the separation between the lines has disappeared under these conditions—the lines are not *resolved*. It is equally evident that the use of a much narrower slit would result in their resolution—the *resolving power* would be increased. In fact, provided the slit width is less than the separation between the lines, the detector output will show a minimum between them.

However, it must be remembered that a narrower slit allows less total energy from the beam to reach the detector and consequently the intrinsic signal strength will be less. There comes a point when decreasing the slit width results in such weak signals that they become indistinguishable from the background noise mentioned in the previous paragraph. Thus spectroscopy is a continual battle to find the minimum slit width consistent with acceptable signal-to-noise values. Improvements in resolving power may arise not only as a result of obtaining better dispersion of the radiation by the analyser but also by using a more sensitive detector.

1.7 THE WIDTH AND INTENSITY OF SPECTRAL TRANSITIONS

In the preceding sections we have seen that a spectral transition has the important property of *position*, measured in terms of its frequency, wavelength, or wavenumber; there are two other important properties, its *width* and its *intensity*, and we shall consider these briefly here.

1.7.1 The Width of Spectral Lines

Throughout this chapter we have drawn spectral absorptions and emissions not as infinitely sharp lines but as more or less broad peaks; we have seen that one reason for this is that the mechanical slits in spectrometers are not infinitely narrow and thus allow a *range* of frequencies, rather than a single frequency, to fall on the detector, hence blurring the pattern. While improvements in spectrometer design can improve the resolving power of an instrument, however, there is nonetheless a minimum width inherent in any atomic or molecular transition—the *natural line width*—beyond which no instrument, however superior, will show a sharpening. This width arises essentially because the energy levels of atomic and molecular systems are not precisely determined, but have a certain fuzziness or imprecision. Several factors contribute to this.

1. *Collision broadening*. Atoms or molecules in liquid and gaseous phases are in continual motion and collide frequently with each other. These collisions inevitably cause some deformation of the particles and hence perturb, to some extent, the energies of at least the outer electrons in each. This immediately gives a possible explanation for the width of visible and ultra-violet spectral lines, since these deal largely with transitions between outer electronic shells. Equally vibrational and rotational spectra are broadened since collisions interfere with these motions too. In general, molecular interactions are more severe in liquids than in gases, and gas-phase spectra usually exhibit sharper lines than those of the corresponding liquid.

   In the case of solids, the motions of the particles are more limited in extent and less random in direction, so that solid-phase spectra are often sharp but show evidence of interactions by the splitting of lines into two or more components.

2. *Doppler broadening*. Again in liquids and gases the motion of the particles causes their absorption and emission frequencies to show a Doppler shift; since the motion is random in a given sample, shifts to both high and low frequencies occur and hence the spectral line is broadened. In general, for liquids collision broadening is the most important factor, whereas

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for gases, where collision broadening is less pronounced, the Doppler effect often determines the natural line width.

3. *Heisenberg uncertainty principle.* Even in an isolated, stationary molecule or atom the energy levels are not infinitely sharp, due to the operation of a fundamental and very important principle, the uncertainty principle of Heisenberg. In effect this says that, if a system exists in an energy state for a limited time δt seconds, then the energy of that state will be uncertain (fuzzy) to an extent δE where

$$\delta E \times \delta t \approx h/2\pi \approx 10^{-34} \text{ J s} \tag{1.10}$$

where h is again Planck's constant. Thus we see that the lowest energy state of a system *is* sharply defined since, left to itself, the system will remain in that state for an infinite time; thus $\delta t = \infty$ and $\delta E = 0$. However, the lifetime of an excited electronic state, for example, is usually only about 10^{-8} s, which gives a value for δE of about $10^{-34}/10^{-8} = 10^{-26}$ J. A transition between this state and the ground state will thus have an energy uncertainty of δE and a corresponding uncertainty in the associated radiation frequency of $\delta E/h$, which we can write as:

$$\delta \nu = \frac{\delta E}{h} \approx \frac{h}{2\pi h \delta t} \approx \frac{1}{2\pi \delta t} \tag{1.11}$$

Thus for our example of an excited electronic state lifetime of 10^{-8} s, $\delta \nu \approx 10^8$ Hz. This apparently large uncertainty is, in fact, small compared with the usual radiation frequency of such transitions, 10^{14}–10^{16} Hz, and so the natural line width is said to be small; in fact, the apparent widths of electronic transitions are far more dependent on collision and Doppler broadening than on energy uncertainties.

On the other hand an excited electron *spin* state may exist for some 10^{-7} s which, from Eq. (1.11), leads to a frequency uncertainty of some 10^7 Hz for a transition. This, compared with the usual frequency of such transitions, 10^8–10^9 Hz, represents a very broad transition indeed, and here the Heisenberg uncertainty relation is by far the most important effect.

Further examples of the application of Heisenberg's principle will be given in later chapters.

1.7.2 The Intensity of Spectral Lines

When discussing spectral intensities there are three main factors to be considered: the likelihood of a system in one state changing to another state—the *transition probability*; the number of atoms or molecules initially in the state from which the transition occurs—the *population*; and the amount of material present giving rise to the spectrum—the *concentration* or *path length* of the sample.

1. *Transition probability.* The detailed calculation of absolute transition probabilities is basically a straightforward matter, but as it involves a knowledge of the precise quantum mechanical wave functions of the two states between which the transition occurs, it can seldom be done with accuracy and is, in any case, beyond the scope of this book. We shall generally content ourselves with qualitative statements about relative transition probabilities without attempting any detailed calculations.

   At a much lower level of sophistication, however, it is often possible to decide whether a particular transition is forbidden or allowed (i.e. whether the transition probability is zero or non-zero). This process is essentially the deduction of *selection rules*, which allow us to decide between which levels transitions will give rise to spectral lines, and it can often be carried out

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through pictorial arguments very like those we have already used in discussing the activity or otherwise of processes in Sec. 1.3.

2. *Population of states.* If we have two levels from which transitions to a third are equally probable, then obviously the most intense spectral line will arise from the level which initially has the greater population. There is a simple statistical rule governing the population of a set of energy levels.

For example, if we have a total of N molecules distributed between two different energy states, a lower and an upper with energies E_{lower} and E_{upper}, respectively, we would intuitively expect most of the molecules to occupy the lower state. Proper statistical analysis bears this out and shows that, *at equilibrium*:

$$\frac{N_{\text{upper}}}{N_{\text{lower}}} = \exp(-\Delta E/kT) \tag{1.12}$$

where $\Delta E = E_{\text{upper}} - E_{\text{lower}}$, T is the temperature in K, and k is a universal constant. The expression is known as the Boltzmann distribution, after its originator, and k, which has a value of $1.38 \times 10^{-23}\ \text{J K}^{-1}$, as Boltzmann's constant. Examples showing the use of this very important expression will recur throughout the remaining chapters.

3. *Path length of sample.* Clearly if a sample is absorbing energy from a beam of radiation, the more sample the beam traverses the more energy will be absorbed from it. We might expect that twice as much sample would give twice the absorption, but a very simple argument shows that this is not so. Consider two identical samples of the same material, S_1 and S_2, and assume that S_1 or S_2 alone absorb 50 per cent of the energy falling on them, allowing the remaining 50 per cent to pass through. If we pass a beam of initial intensity I_0 through S_1, 50 per cent of I_0 will be absorbed and the intensity of the beam leaving S_1 will be $\frac{1}{2}I_0$; if we then pass this beam through S_2 a further 50 per cent will be absorbed, and $\frac{1}{2} \times \frac{1}{2}I_0 = \frac{1}{4}I_0$ will leave S_2. Thus two 50 per cent absorptions in succession do not add up to 100 per cent but only to 75 per cent absorption. An exactly similar relationship exists between the *concentration* of a sample and the amount of energy absorption—a doubling of the concentration produces something less than a doubling of the absorption.

The relationship between concentration (c), path length (l), and the incident and transmitted intensities of radiation, (I_0 and I, respectively) can be expressed in many ways, all based on the *Beer–Lambert law*, which is often written:

$$I/I_0 = \exp(-\kappa c l) \tag{1.13a}$$

where κ is a constant for the particular spectroscopic transition under investigation. To remove the inconvenience of using an exponential function, Eq. (1.13*a*) may be recast as:

$$I/I_0 = 10^{-\epsilon c l} = T \tag{1.13b}$$

where we have introduced the symbol T for *transmittance*, defined simply as the ratio I/I_0; ϵ is called the *molar absorption coefficient*, which is the term now favoured by the International Union of Pure and Applied Chemistry (IUPAC), although other terms still common are the *decadic extinction coefficient*, or *decadic absorptivity* (*decadic* meaning related to base 10 rather than base e).

When spectroscopy is used to measure concentration of material it is convenient to have a relationship which, unlike Eqs (1.13*a*) or (1.13*b*), is linear in concentration. Inverting Eq. (1.13*b*) and taking logarithms we have:

$$I_0/I = 10^{\epsilon c l} \quad \text{or} \quad \log(I_0/I) = \epsilon c l = A \tag{1.13c}$$

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where we define the important quantity A, called the *absorbance*, or *optical density*. A is directly proportional to concentration and, because of its convenience in this respect, some spectrometers are constructed to record spectra directly in absorbance units.

1.8 FOURIER TRANSFORM SPECTROSCOPY

One of the major disadvantages of the conventional method of producing a spectrum such as that of Fig. 1.10(*b*) is its inherent slowness. Each point of the spectrum has to be recorded separately—the spectrometer is set to start reading at one end, the frequency is swept smoothly across the whole span of the spectrum, and the detector signal is monitored and recorded. The inefficiency of such a method is clear when one considers taking a spectrum with only one or two peaks in it; we have to sweep from one end to the other in order to find the peaks, but most of the time is spent recording nothing but background noise. Initially it was only in the visible and ultra-violet regions that the whole of a spectrum could be recorded simultaneously (on a photographic plate), but the development of Fourier transform (FT) spectroscopy now provides simultaneous and almost instantaneous recording of the whole spectrum in the magnetic resonance, microwave and infra-red regions. In this section we shall briefly discuss the basic ideas of the technique, leaving to later chapters more detailed consideration of its methods and applications.

Although equally applicable to both emission and absorption spectroscopy, it is easier to visualize Fourier transform spectroscopy in terms of emission. For the moment, too, we shall ignore the line-broadening discussed in the previous section and think of the emitted radiation as a pure cosine wave at some quite precise frequency, ν. If a detector capable of responding sufficiently rapidly receives this emitted radiation, its output will be an oscillating signal, again of frequency ν. Note carefully that here we think of the detector output as a function of *time* ('time domain spectroscopy') rather than as the function of frequency ('frequency domain') previously considered.

Now imagine a sample emitting radiation at two different frequencies; a detector receiving the total radiation will 'see' the *sum* of the two cosine waves. We illustrate, diagrammatically, two separate but superimposed waves in Fig. 1.15(*a*), where the lower frequency wave is shown dashed, and their sum in Fig. 1.15(*b*); clearly the detector output, plotted against time, shows both an oscillation due to the frequency of the two waves, and a slow increase and decrease in overall amplitude. The frequency of this latter oscillation is often called the 'beat' frequency, by analogy with a similar phenomenon for musical tones, and it arises because the two component waves are sometimes in step, where they totally reinforce each other (points labelled A in the figure), and sometimes out of step, where they cancel (points B).

If we *decrease* the difference in frequency between the component waves, they will get out of step more slowly and it takes longer for them to get back totally in step again. This is shown in Fig. 1.15(*c*) and (*d*), with the points marked A and B indicating, as before, where the waves are in step and totally out of step, respectively. The separate waves are drawn in Fig. 1.15(*c*) and their sum in Fig. 1.15(*d*), and we note that the beat frequency is lower than in Fig. 1.15(*b*). In general, it is easily shown that the beat frequency is equal to the difference in frequency of the two component waves.

Mathematically it is simple, although tedious, to resolve a combined wave such as Fig. 1.15(*b*) or (*d*) into its components. Essentially each component wave has its own frequency and maximum amplitude, so two components require the evaluation of four unknowns from the composite curve. In principle, then, observations of the time domain signal at four points and solution of four simultaneous equations will yield the information we seek.

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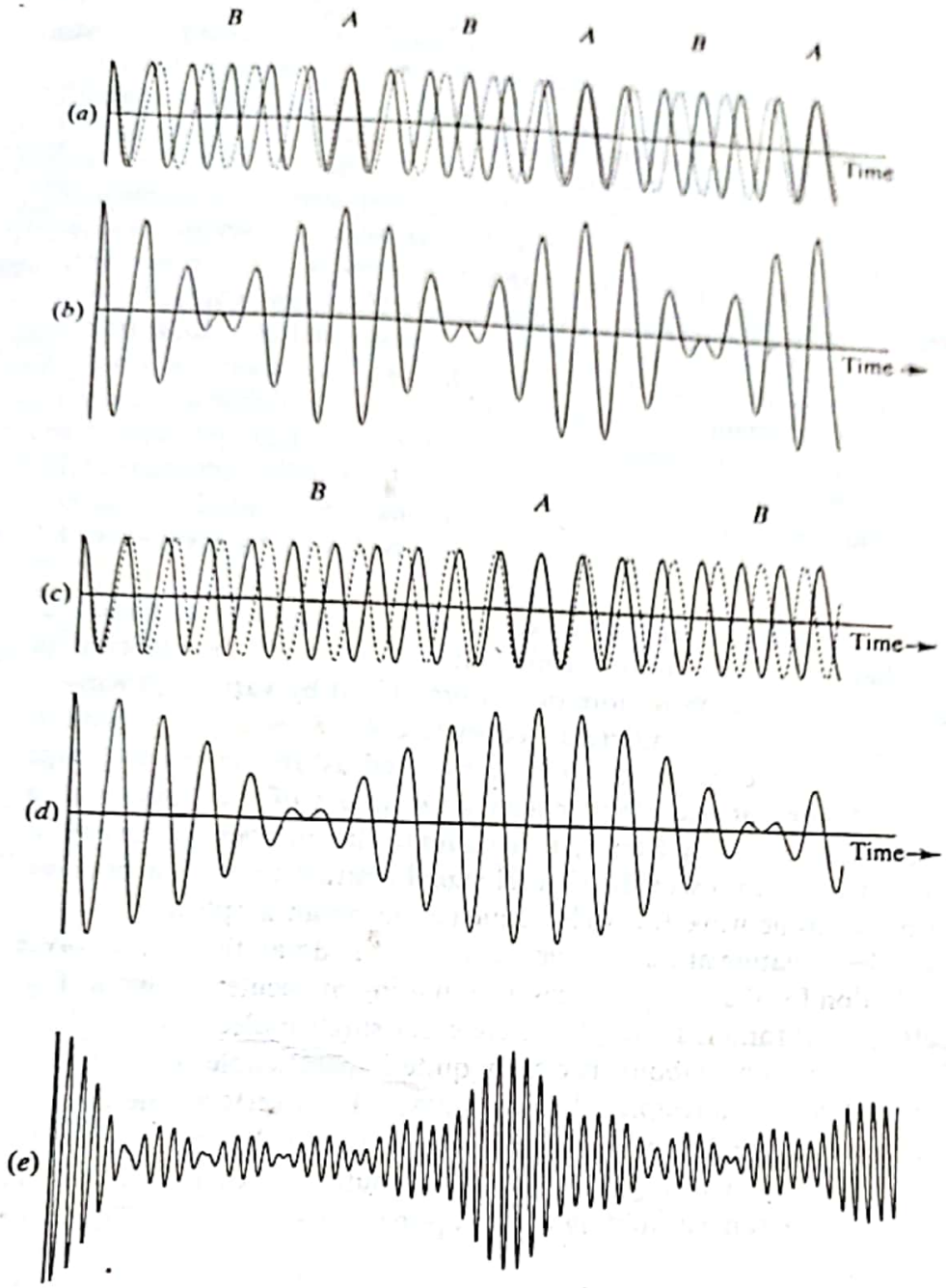


Figure 1.15 Adding cosine waves: (*a*) and (*c*) show the superposition of two cosine waves with slightly different frequencies; (*b*) and (*d*) show their sums. The summation of six cosine waves with different frequencies is shown in (*e*).

Adding more than two cosine waves complicates the resultant combined wave and makes the resolution into components even more tedious, but does not change the principle. Figure 1.15(*e*) shows the result of superimposing six cosine waves of different frequencies. It would need 12 measured points and the solution of 12 simultaneous equations to determine the frequency and relative amplitude of each component. Fortunately there is a simple and quite general way to resolve a complex wave into its frequency components; this is the mathematical process known as Fourier transformation, named after the French mathematician Jean Baptiste Fourier, who developed the method in the early 1800s. Even more fortunately we do not need to know how the process works; it suffices to say that it is essentially a matter of integration of the complex waveform and that it is now carried out very conveniently by computer.

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As an example of its operation let us imagine that a suitable detector is responding to the complex waveform of Fig. 1.15(*b*). A computer receiving the detector output might typically be set to sample it once every millisecond and to store, say, 2000 samplings in separate memory locations; it would thus need to collect the signal for just two seconds. The computer would then apply the Fourier transform process to the stored data, taking a further second or so, and the component cosine wave frequencies and amplitudes could be displayed. Conventionally the display would not take the form of Fig. 1.15(*a*), where the actual periodic variation of the waves is shown, but would instead be the *spectrum* of the waves—i.e. two very sharp peaks of equal height plotted on a suitable frequency scale to show where the two frequencies occur. This is shown in Fig. 1.16, where the complex wave in Fig. 1.16(*a*) (taken from Fig. 1.15(*b*)) is seen to give rise to the spectrum of Fig. 1.16(*b*). Essentially the Fourier transform has converted the *time* domain plot of Fig. 1.16(*a*) into the *frequency* domain spectrum of Fig. 1.16(*b*). The process described above would have taken, perhaps, five seconds only. The detector collects *all* the spectral information virtually simultaneously and the computer 'decodes' that information into the conventional spectrum. It is in this way that the FT method speeds the collection of spectral data, typically, by factors of 10 to 1000.

We must consider one or two more points before we leave this discussion of basic FT spectroscopy. Firstly, recall that real samples do not emit radiation at precise frequencies; as we saw in the previous section, each emission is more or less broadened by various processes, and so each 'line' is really a package of slightly different frequencies. We show a typical peak in Fig. 1.17(*a*). In Fig. 1.17(*b*) we see that the peak can be considered as arising from a large number of sample molecules radiating at ν_{max}, the frequency maximum of the peak, with a smaller number radiating at frequencies away from that maximum, the number decreasing as the separation increases. If we wanted to discover the total signal emitted by such a peak we could, if we had time, plot out a cosine wave for each frequency, using an amplitude proportional to the number of molecules radiating at that frequency, and then add all the cosine waves together. To carry out this addition for the few 'packages' of radiating molecules shown in Fig. 1.17(*b*) would not be difficult; to do it for a real sample, where every single molecule might have a radiation frequency different from its neighbour, is clearly quite impracticable. Fortunately, however, the Fourier transform process is a reciprocal one—just as FT converts a time domain signal to a frequency domain spectrum (e.g. Fig. 1.16), so it will carry out the reverse conversion. Thus if we supply the frequency curve of Fig. 1.17(*a*) to a computer and carry out the FT, the resultant display will be exactly the same as adding the component cosine waves. The result is shown in Fig. 1.17(*c*).

We see that a detector receiving the total radiation from a single broad-line emission will show an oscillating signal whose overall amplitude decays smoothly to zero. The oscillation is the beat pattern set up by all the superimposed, but slightly different, cosine waves; the signal

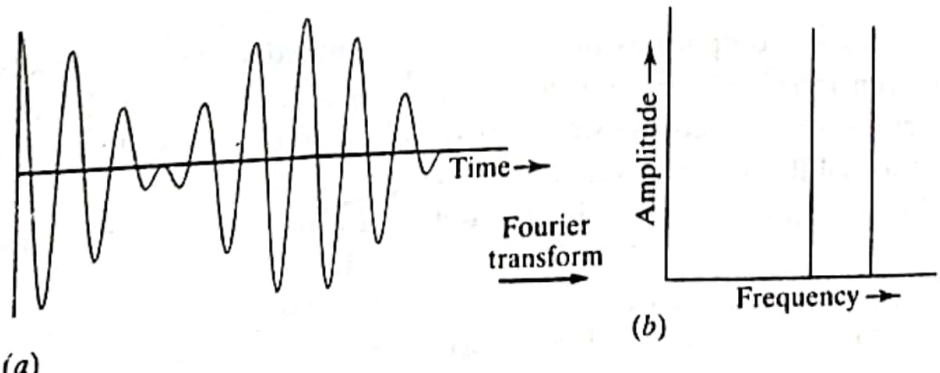


Figure 1.16 The use of the Fourier transform to convert the summed cosine waves of (*a*) (redrawn from Fig. 1.15(*b*)) to the frequency spectrum of (*b*).

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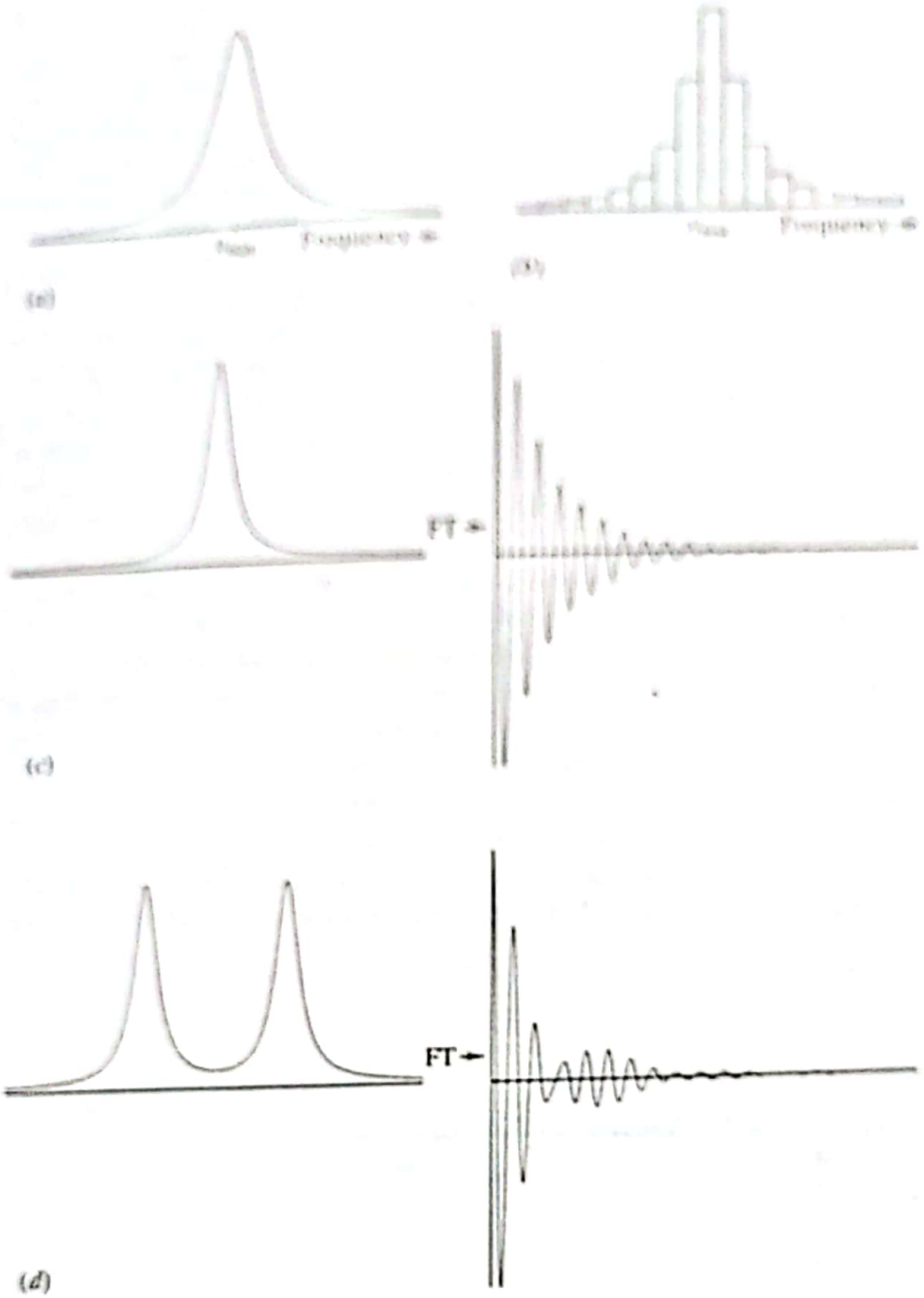


Figure 1.17 (*a*) The frequency distribution of a broad spectral line and (*b*) the histogram of its frequency distribution. The Fourier transform of (*c*) a single broad spectral line and (*d*) a pair of broad lines.

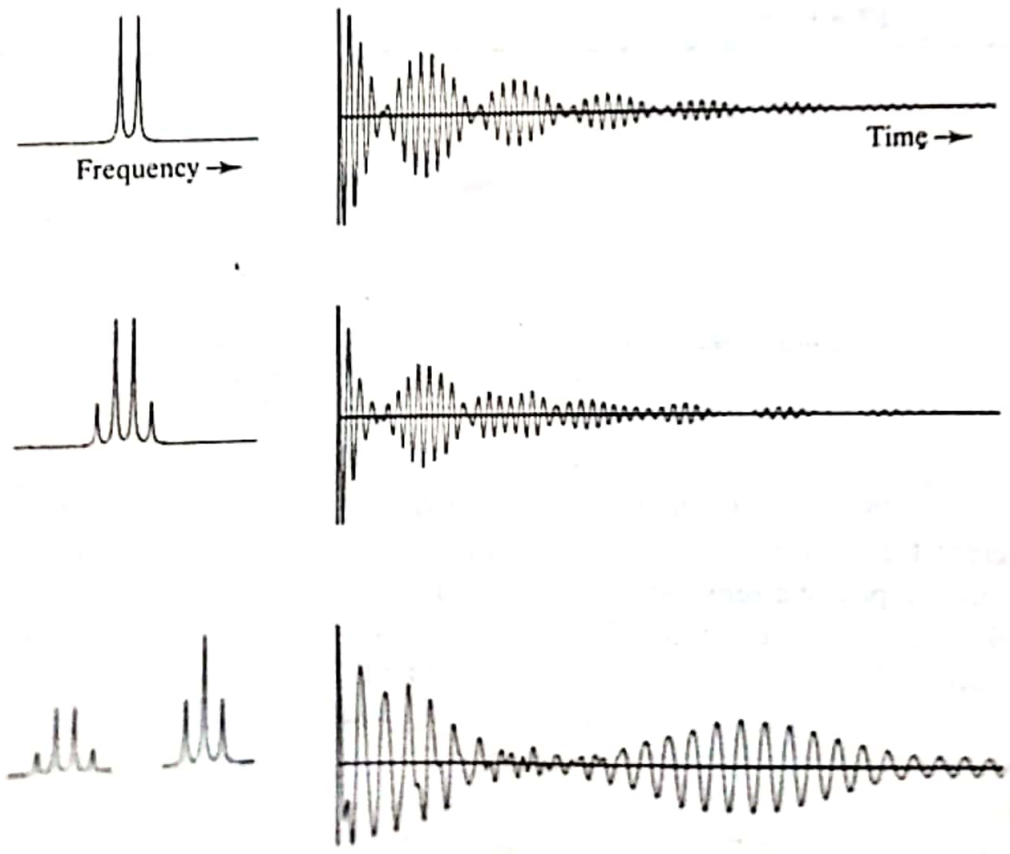
decays because, if we imagine all the waves in the peak to be 'in step' initially, after some time has elapsed the many different frequencies will be very much out of step, and on average half will have their amplitudes in the positive sense and half in the negative, giving a resultant of zero. Another way to think of this is to remember that two waves setting out in step with an infinitesimally small difference in frequency will take an infinite time to get back in step again, i.e. they will never do so. The frequency distribution of Fig. 1.17(*a*) has many infinitely close frequencies within it and so, after a few cycles, none of the individual waves ever get back into step again. If the band had been *infinitely* broad, i.e. containing an infinite number of infinitesimally close neighbours, none would ever have been in step after the first instant, and the FT of such a 'white' source is a single decaying signal with no beats. We shall return to this in a moment.

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The corollary of these arguments is that the *rate of decay* of the overall signal depends on the *width* of the original frequency peak. A broader peak contains a larger number of different frequencies and so will decay more rapidly. Clearly both the position and the width of a frequency peak can be recovered from the time domain signal.

Next we should briefly consider the situation when a sample can undergo more than one spectroscopic transition, so that its frequency spectrum shows more than one peak. As before, the overall time domain signal is the sum of the various beat patterns set up by adding cosine waves from each broad peak. We show some typical patterns in Fig. 1.18, all taken from the type of frequency spectra typically found in nuclear magnetic resonance spectroscopy where the FT technique has found particularly wide application. Clearly the complexity of the time domain signal increases as more peaks are added; the point to remember is that all the information regarding the position, intensity, and width of each spectral line on the left of the figure is contained in the time domain pattern on the right; after experimentally observing the latter directly for a brief period, the frequency spectrum can be quickly recovered by Fourier transformation.

Although we stated initially that the FT process is most easily visualized in terms of the emission of radiation, the technique is just as readily applied to absorption. We have already seen that a 'white' source would show a single decay signal with no beats; an approximation to this is shown in Fig. 1.19(*a*), where a *very* broad emission line (which can be considered as a white source covering a limited region of the spectrum) and its Fourier transform are shown. Although the time domain signal decays very rapidly, it does show some beats; this is because the broad emission line is only an approximation to a proper white source. We can now imagine an absorbing sample making some 'holes' in this radiation, as shown on the left-hand side of Fig. 1.19(*b*), with its resulting FT signal on the right. Although we may find it difficult to imagine Fourier-transforming (or even just adding up) the *absence* of radiation at some



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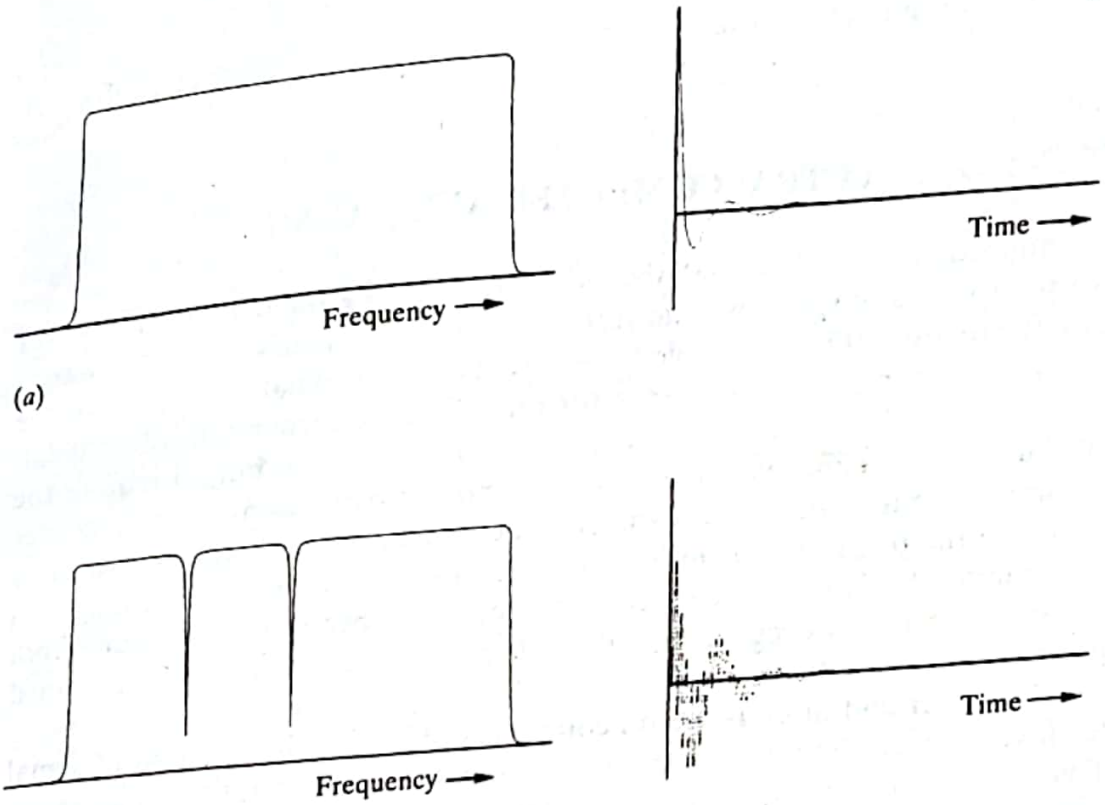


Figure 1.19 (*a*) An approximation to a 'white' source and its Fourier transform. (*b*) Some absorptions from a 'white' source and their Fourier transform.

frequencies, in practice a detector will collect a perfectly sensible signal which can be stored by a computer, transformed, and displayed as the normal absorption spectrum.

Finally we should point out that, although this introduction to the FT process has concentrated on *time* domain signals, monitoring the detector output as a function of time is not the only way in which the total spectrum from a sample can be collected. The student of physics will have recognized that the patterns of Fig. 1.15 can equally be described in terms of interference phenomena. In Chapter 3 we shall find that FT spectroscopy in the infra-red region operates by observing the interference pattern set up when the radiation beam is reflected between a pair of parallel mirrors which are steadily moved towards or away from each other; it is this pattern which is collected, recorded as a function of *distance* between mirrors, and later Fourier-transformed into the corresponding frequency spectrum. Whether to use interferometry or time domain signals in a particular application is based on purely practical grounds; it is difficult to find an infra-red detector with a response time sufficiently fast to be able to record a time domain signal, but it is very easy to reflect infra-red radiation between mirrors. Conversely, in the radiofrequency region, fast response detectors are no problem, but it is not at all easy to reflect the radiation between 'mirrors' in a sensible and coherent fashion.

We see then that the FT method allows us to record spectra much more rapidly than the conventional frequency sweep technique. This in itself is valuable; spectrometers are costly instruments, and the more work we can get from them in a given time the more justified the initial investment. However, rapid data collection brings other benefits, e.g. in being able to record the spectra of transient species such as unstable molecules or intermediates in a chemical reaction. Since the technique essentially reduces the time spent obtaining a spectrum from hours or minutes down to seconds or even fractions of a second, it vastly increases the range of materials which can be studied. There are other advantages of using FT instruments, but we

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shall leave discussion of them until the relevant chapters on magnetic resonance and infra-red spectroscopy.

1.9 ENHANCEMENT OF SPECTRA: COMPUTER AVERAGING

We have already mentioned, in Sec. 1.6, that the problem of background noise imposes a limitation on the sensitivity of any spectroscopic technique—unless a real signal peak stands out clearly from noise fluctuations it is impossible to be sure that it *is* a signal. A signal-to-noise (S/N) ratio of 3 or 4 is usually reckoned necessary for unambiguous recognition of a signal. There are several ways in which S/N can be improved for a given sample, but all require the expenditure of time. Thus it is possible electronically to damp out oscillations of the recorder pen so that it is less susceptible to high-frequency noise. The baseline of the spectrum will then be smoother, but, because the pen responds more slowly to any change (including changes in signal), one must sweep more slowly across the spectrum. Nor is FT spectroscopy immune from noise—detector and amplifier noises occur during the collection of data and are transformed into spurious frequencies in the spectrum.

The availability of cheaper and more powerful computers offers another method of signal enhancement, which involves recording the spectrum stepwise into a computer. Of course, this is already done if FT is intended, but it is just as easy to sample a frequency domain spectrum at, say, 2000 closely spaced points, and to store the intensity at each point in 2000 separate computer memory locations. This process may then be repeated as many times as we wish, but each time *adding* the new data into that already existing. Although in any one scan a weak signal may not be visible above the noise level, after n summed scans the signal will be n times larger in the store, whereas the noise, being random, will sometimes contribute to the store in a positive sense and sometimes negatively, so it will accumulate less rapidly. In fact it may be shown that n scans increase the noise level in the store by $n^{1/2}$, so the net gain in S/N is $n/n^{1/2} = n^{1/2}$.

If a single scan takes several minutes, as is usually the case in conventional frequency-sweep spectroscopy, the necessity to store 100 scans in order to give an improvement in S/N by a factor of 10 is rather costly in instrument time, so computer averaging is not often used in such measurements. However, the combination of computer averaging with Fourier transform is very powerful indeed. Here one time domain scan can be completed in a second or two, and 100 scans will only occupy a couple of minutes; thus a tenfold gain in S/N can be achieved in a total time often less than that required for a non-enhanced spectrum by ordinary sweep methods.

Other benefits follow from the addition of a computer to a spectrometer. The spectrum of a solvent or other background can be stored in the computer and subtracted from the observed spectrum in order to isolate the spectrum of the substance, or peak intensities can be automatically measured and converted to sample concentrations. Even the operation of the spectrometer itself can usefully be entrusted to the computer—samples can be changed automatically, and the optimum operating conditions can be determined and set for each new sample.

1.10 STIMULATED EMISSION: LASERS

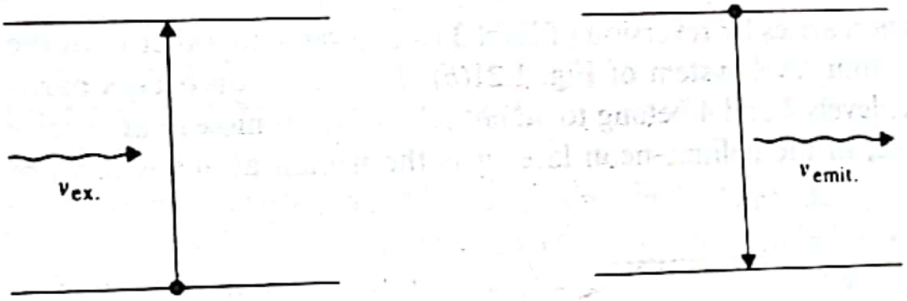
We have already mentioned that, once radiation has been absorbed by a sample, the sample can lose its excess energy either by thermal collisions or by re-emission of radiation. In this section we shall consider the latter process in more detail, because it leads to the very important topic of laser radiation.

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Radiation may be emitted by an excited molecule or atom either *spontaneously* or as the result of some stimulus acting on the molecule, called *stimulated emission*. Which of these two processes is most likely to occur in any given case depends on the energy jump involved, i.e. on the frequency of the radiation being emitted. For high-frequency transitions (infra-red, visible, and ultra-violet upwards) spontaneous emission is by far the most likely; conversely, for low-frequency changes (microwave and magnetic resonance) spontaneous emission is unlikely and, if the right conditions obtain, stimulated emission will occur.

Stimulated emission is a resonance phenomenon—an excited state drops to the ground state (emitting radiation of frequency $\nu = \Delta E/h$, where ΔE is the energy gap) only when a photon (i.e. radiation) of the same frequency ν interacts with the system. We illustrate the situation in Fig. 1.20. On the left, in both (*a*) and (*b*), we show the excitation of a molecule by absorption of radiation of frequency $\nu_{ex.}$. At the right in (*a*) we show spontaneous emission, when radiation $\nu_{emit.}$ is spontaneously given out, and in (*b*) stimulated emission where a photon of frequency $\nu_{stim.}$ interacts with the excited state and causes radiation of frequency $\nu_{emit.}$ to be released. Note particularly that, although we have given different subscripts to $\nu_{ex.}$, $\nu_{emit.}$, and $\nu_{stim.}$ in order to indicate their origins, they all represent *exactly the same frequency*, the frequency $\Delta E/h$.

Radiation emitted under stimulation of this sort has three very important qualities. Firstly, it is of a very *precisely defined* frequency: the excited state does not spontaneously decay, so it is inherently long-lived, which implies (see the discussion of Heisenberg uncertainty in Sec. 1.7.1) a narrow energy level. Secondly, the emitted radiation is *in phase* with the stimulating radiation: the excited state is stimulated to emit by interaction with the oscillating electromagnetic field of $\nu_{stim.}$, so it is not surprising that the maximum amplitude of the emitted wave coincides with that of $\nu_{stim.}$. Since the waves are of exactly the same frequency, they remain in phase as they leave the sample. Finally, the stimulating and emitted radiation are *coherent*, which means that they travel in precisely the same direction. In contrast *spontaneous* emission can occur at any time (so each emitted photon is not necessarily in phase with any other), in any direction, and within a more or less broad range of frequencies.



(*a*)



(*b*)

Figure 1.20 Showing (*a*) spontaneous and (*b*) stimulated emission from an excited energy state.

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Of course the stimulating radiation of Fig. 1.20 is still present in the system after emission has occurred—it is in no way absorbed—so it can go on to interact with another excited molecule to induce more emission. Equally the emitted radiation has the right frequency to stimulate emission from yet another excited molecule. Clearly, all the time a supply of excited molecules exists, this process is likely to cascade and a great deal of radiation may be emitted coherently. This *amplification* of the original stimulating photon is reflected in the name of the process—*l*ight *a*mplification by *s*timulated *e*mission of *r*adiation, or *laser*.

In fact, as we have said, light (or, more properly, visible radiation) is far more likely to be emitted *spontaneously*, and so not to have the coherent properties of laser radiation. It was in the microwave region that the first successful amplification by stimulated emission was performed (and the process was therefore christened *maser*, standing for *m*icrowave *a*mplification by *s*timulated *e*mission of *r*adiation). For the process to be possible in higher-frequency regions it is necessary to find systems with long-lived excited states so that stimulated, rather than spontaneous, emission may predominate, and this may only be achieved if more than two energy levels are involved.

Consider the three energy levels of Fig. 1.21(*a*). Excitation from the ground state, level 1, to the normal excited state, level 2, can occur by absorption of radiation, as usual. Provided that, as well as emitting spontaneously, level 2 can transfer some molecules into a metastable state (level 3) which cannot easily revert spontaneously to the ground state, then the population of level 3 builds up, and laser action becomes feasible. The ruby laser is an example of this type of three-level system. Ruby is basically aluminium oxide containing a trace (about 0.05 per cent) of chromium ions, which gives it its characteristic colour. A discharge tube wound round a rod of ruby is flashed very briefly to raise the chromium ions into an excited electronic state; they drop rapidly by thermal, non-radiative processes, into a metastable state some 14 000 cm^{-1} above the ground state, and they then revert to the ground state, by laser action initiated by one of the infrequent spontaneously emitted photons emitting radiation at about 690 nm wavelength. The decay from excited to metastable state releases quite large amounts of heat, so the ruby must be allowed to cool before another excitation cycle is commenced; it is thus operated as a pulsed laser.

In some cases the laser emission arises by reversion of level 3 to a lower state other than the original level 1, as shown in the four-level system of Fig. 1.21(*b*). This situation occurs particularly when, as is quite possible, levels 3 and 4 belong to an entirely *different* molecular species from levels 1 and 2. For example, in the helium–neon laser it is the helium atoms which are

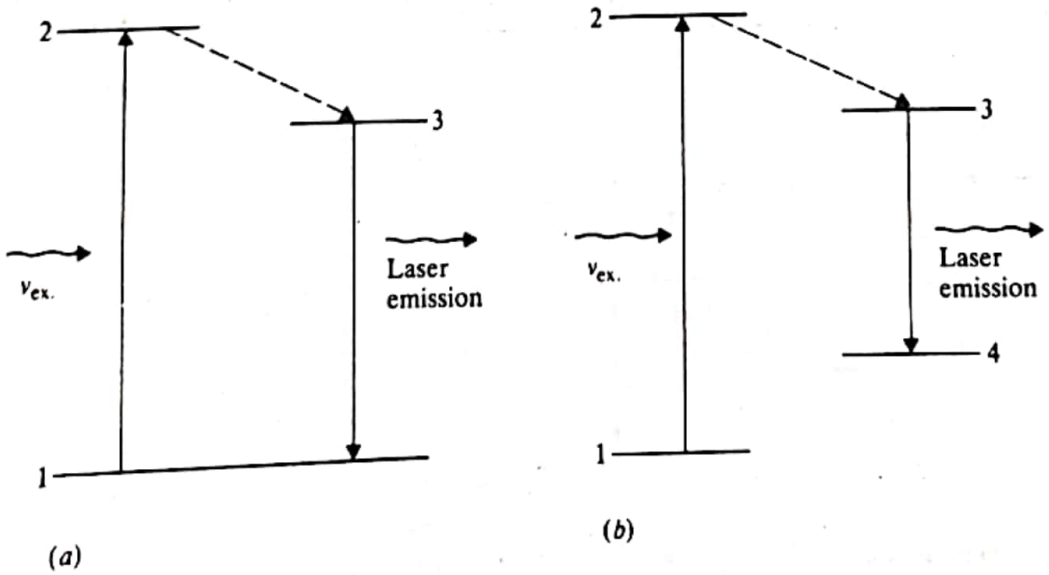


Figure 1.21 The energy levels of (*a*) a three-level and (*b*) a four-level system operating as a laser.

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initially excited (level 1 to 2), and which then transfer their excitation energy to neon atoms by collisions; this can happen only because neon has an excited state with almost exactly the same energy as the excited state of helium, so a resonance transfer of energy is possible. It also happens that the excited state of neon does not readily undergo an ordinary spectroscopic transition back to its ground state, so the conditions are ideal for laser action. Provided the exciting radiation for helium is maintained, so replenishing the population of excited helium atoms, this type of laser can operate continuously. We shall discuss the precise electronic energy levels involved in more detail in Chapter 5.

The extreme coherence of laser radiation makes it ideal in applications like communications or distance measurement. From the spectroscopic point of view it is becoming increasingly important as new types of spectroscopy are developed but, for the purposes of this book, its most far-reaching application is in the area of vibrational spectroscopy. Raman spectroscopy, discussed at length in Chapter 4, requires an intense monochromatic source for which a laser in the visible or near-infra-red region is ideal.

For virtually all other spectroscopic measurements, however, either a wide-band or a tunable source is desirable. Lasers cannot be wide-band (although the CO_2 laser, to be described in Sec. 3.8.4, comes close to this since it gives a series of closely spaced lasing bands), but recent developments have led to their becoming tunable. To change the emission frequency of a laser it is necessary to be able to modify the energy levels between which laser transitions take place. Solids emitting laser radiation can be subjected to varying temperatures or pressures in order smoothly to change the relevant energies, but the extent of such changes is relatively small. More usefully, lasers made from coloured organic substances in solution—the so-called dye lasers—are widely tunable. In these the active material is usually a rare earth ion held in the centre of an organic 'ligand' which complexes firmly to the ion. The system can be tuned, within a continuum corresponding to a broad region of fluorescence, by building a spectrum dispersing element into the laser cavity, and it is now possible to produce laser emission at any frequency from the near-infra-red to the ultra-violet. Such intense, sharply defined but variable-frequency sources will certainly find application in many areas of spectroscopy.

1.11 SYNCHROTRON RADIATION

As spectroscopic measurements become increasingly demanding, means to overcome sensitivity problems are developed and adapted. Sensitivity can often be improved by using a very powerful radiation source to increase the signal-to-noise ratio and so greatly reduce scanning time. Synchrotron radiation, first observed in the late 1940s at the GEC synchrotron at Schenectady, New York, is such a source, and it has the huge additional advantage of being readily tunable. Since 1940 a number of synchrotrons have been constructed in various locations around the world, and the advantages to spectroscopists are such that there is considerable competition for 'beamtime'.

An electron synchrotron consists of a large ring-shaped vacuum chamber into which pulses of high-velocity electrons are injected after acceleration by electric fields. Magnetic fields, provided by magnets arranged around the ring, force the electrons to follow the curved path. At velocities close to that of light, electrons emit radiation when they are accelerated so that a narrow beam of radiation is produced tangentially to their path.

This so-called synchrotron radiation can be tuned over a wide range of frequencies by changing the accelerating magnetic field. A number of different bending magnets (the two main types of which are called 'wigglers' and 'undulators') can be used on a single synchrotron ring to deliver radiation of a particular frequency range via a beam line to a spectrometer placed

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outside the ring. The tunability and brilliance of these sources enable many otherwise extremely difficult experiments to be carried out. Some newer experimental techniques, such as EXAFS, that are described in Chapter 8 have been developed using synchrotron sources.

BIBLIOGRAPHY

George, W. O., and H. O. Willis: *Computer Methods in Ultraviolet, Visible and Infra-red Spectroscopy*, Royal Society of Chemistry, 1990

PROBLEMS

(Useful constants: $N = 6.023 \times 10^{23}\,\text{mol}^{-1}$; $k = 1.381 \times 10^{-23}\,\text{J K}^{-1}$; $h = 6.626 \times 10^{-34}\,\text{J s}$; $c = 2.998 \times 10^{8}\,\text{m s}^{-1}$.)

1.1 Convert the following spectroscopic quantities as indicated:

$2000\,\text{cm}^{-1}$	to μm
0.15 nm	to Hz
500 nm	to cm^{-1}
9 GHz	to cm^{-1}

State in which region of the electromagnetic spectrum you would expect each to appear, and what sort of transition each corresponds to.

1.2 The wavelength of the radiation absorbed during a particular spectroscopic transition is observed to be 10 μm. Express this in frequency (Hz) and in wavenumber (cm^{-1}), and calculate the energy change during the transition in both joules per molecule and joules per mole. If the energy change were twice as large, what would be the wavelength of the corresponding radiation?

1.3 Which of the following molecules would show (*a*) a microwave (rotational) spectrum, (*b*) an infra-red (vibrational) spectrum: Br_2, HBr, CS_2?

1.4 How would you expect widening of the slits in a spectrometer to affect its resolution and its sensitivity?

1.5 State the Beer–Lambert law. The transmittance of an aqueous solution of $KMnO_4$ at a certain wavelength is 1 per cent (that is 0.01) for a 10^{-3} molar solution in a 1 cm cell. What is (*a*) its absorbance and (*b*) the molar absorption coefficient of $KMnO_4$?

1.6 A particular molecule is known to undergo spectroscopic transitions between the ground state and two excited states, (*a*) and (*b*), its lifetime in (*a*) being about 10 s and in (*b*) about 0.1 s. Calculate the approximate uncertainty in the excited state energy levels and the widths of the associated spectral 'lines' in hertz.

1.7 A certain transition involves an energy change of $4.005 \times 10^{-22}\,\text{J molecule}^{-1}$. If there are 1000 molecules in the ground state, what is the approximate equilibrium population of the excited state at temperatures of (*a*) 29 K, (*b*) 290 K, and (*c*) 2900 K? What would your answer have been if the energy change were 10 times greater?

1.8 Sketch the curve you would expect from Fourier transformation of a sine wave.

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